`bispectrum`: Selective G-Bispectra Made Practical

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Abstract

We present `bispectrum`, an open-source PyTorch library that computes selective G-bispectral invariants for seven group/domain pairs as drop-in differentiable `nn.Modules`. The selective bispectrum reduces invariant cost from $O(|G|^2)$ to $O(|G|)$ while remaining complete, i.e. it separates orbits for generic signals. Building the library required two theoretical contributions: (i) an augmented selective $\mathrm{SO}(3)$ invariant on S^2—combining $O(L^2)$ scalar bispectral entries with Clebsch–Gordan power spectrum entries needed for real signals—that achieves $\Theta(L^2)$ coefficients, matching the information-theoretic lower bound, instead of $O(L^3)$ for the full bispectrum (completeness proved for $L \leq 5$; the construction extends to any L), and (ii) resolving a previously undocumented inversion failure for the octahedral group via bootstrap initialization followed by Levenberg–Marquardt refinement. We evaluate the library on three benchmarks—C_8 on PCam, octahedral on OrganMNIST3D, and $\mathrm{SO}(3)$ on Spherical MNIST—demonstrating numerically exact rotation invariance ($\sigma_{\mathrm{rot}} < 10^{-3}$) and a data-efficiency advantage of complete bispectral invariants over incomplete alternatives in the low-data, moderate-capacity regime, where the complete invariant map substitutes for missing training examples. The selective forward pass runs in sub-millisecond time on GPU for finite groups, adding negligible overhead to standard training loops.

1 Introduction

Equivariant neural networks [Cohen and Welling, 2016, Weiler and Cesa, 2019] enforce symmetry by construction, but they still require an invariant representation at the point where a prediction must no longer depend on orientation. In practice, invariant pooling remains a weak point: standard choices such as norm pooling, gate-based pooling, or max/sum aggregation are usable [Weiler and Cesa, 2019], but they do not preserve all equivariant information. The bispectrum offers a mathematically principled alternative, since for many groups it yields exact invariants with reconstruction guarantees [Kakarala, 1992].

The obstacle is practical rather than conceptual. Existing bispectral constructions are scattered across theory papers, domain-specific derivations, and small implementations, with no general-purpose PyTorch library covering the finite-group and compact-group cases relevant to modern geometric deep learning. Even for groups where selectivity reduces the coefficient count from $O(|G|^2)$ to $O(|G|)$ [Mataigne et al., 2024], a user still needs a numerically stable implementation, a consistent API, inversion routines, tests, and benchmarked runtime behavior.

This paper presents `bispectrum`, a PyTorch [Paszke et al., 2019] library for computing bispectral invariants and, where available, their inverses. The library supports cyclic, dihedral, torus, disk, sphere, and octahedral settings as differentiable `nn.Modules` intended to plug into standard ML workflows. Beyond the software contribution, the paper makes two technical contributions that arose during implementation: an augmented selective $\mathrm{SO}(3)$ bispectrum on S^2 with $O(L^2)$ coefficients

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Table 1: Invariant nonlinearities for equivariant CNNs. The selective bispectrum is the only method that is simultaneously exact, complete, and linear-cost for finite groups.

Nonlinearity	Exact?	Complete?	Coefficients	Key limitation
Norm	Yes	No	$O(\|G\|)$	Discards phase
Gate	Yes	No	$O(2\|G\|)$	Scalar bottleneck
Fourier pointwise	No	Yes	$O(\|G\| \log \|G\|)$	Aliasing
Full bispectrum	Yes	Yes	$O(\|G\|^2)$	Quadratic cost
Sel. bispectrum	**Yes**	**Yes**	$\mathbf{O(\|G\|)}$	**(this work)**

(Section 3.4), and a resolution of the bootstrap inversion failure [Mataigne et al., 2024] for the octahedral group (Section 3.5).

Contributions.

1. **Software contribution.** A PyTorch library implementing bispectral invariants for seven group/domain pairs with a consistent `nn.Module`-style interface, GPU-optimised forward passes (<1 ms for finite groups), precomputed Clebsch–Gordan buffers, and parallel construction routines (Section 3.6).
2. **Augmented selective $\mathrm{SO}(3)$ bispectrum on S^2.** We construct and implement an augmented selective bispectrum for $\mathrm{SO}(3)$ acting on S^2 that uses $\Theta(L^2)$ coefficients instead of the full $\Theta(L^3)$, matching the orbit-space lower bound up to a constant factor. The invariant combines scalar bispectral entries with Clebsch–Gordan power spectrum entries (a degree-4 complement needed for real signals). Completeness is proved for $L \leq 5$ via certified interval arithmetic; the construction extends to any L.
3. **Octahedral inversion.** An analysis of why the standard selective-bispectrum bootstrap inversion fails for the octahedral group, together with a working bootstrap + Levenberg–Marquardt correction.
4. **Empirical validation.** A task-driven evaluation of the library as an invariant pooling primitive across three group/domain settings, showing that complete bispectral invariants provide a data-efficiency advantage in the low-data, moderate-capacity regime while converging to incomplete alternatives at high capacity.

2 Mathematical Background

2.1 Equivariant features and invariant maps

A neural network layer is *G-equivariant* if its output transforms predictably under the group action: $[\pi_{\mathrm{out}}(g) \circ \Phi](f) = \Phi(\pi_{\mathrm{in}}(g) f)$ for all $g \in G$ [Cohen and Welling, 2016, Weiler and Cesa, 2019]. Stacking equivariant layers preserves this property, but any task that requires a prediction independent of orientation—classification, regression, detection—must eventually map equivariant features to *invariant* ones.

Table 1 summarises the trade-offs. Norm pooling and gated nonlinearities are exact invariants but *incomplete*: they discard the phase of Fourier blocks, collapsing distinct orbits. Fourier-space pointwise nonlinearities preserve completeness but sacrifice exactness through aliasing [Weiler and Cesa, 2019]. The full bispectrum is exact and complete but costs $O(|G|^2)$ coefficients.

2.2 The G-bispectrum and selective bispectra

Let G be a compact group with irreducible representations $\{\rho_i\}_{i \in \hat{G}}$ of dimensions d_i, and let $\hat{f}(\rho_i) = \sum_{g \in G} f(g)\, \rho_i(g)$ denote the group Fourier transform. Given the Clebsch–Gordan decomposition $\rho_i \otimes \rho_j \cong \bigoplus_k m_{ij}^k \rho_k$ with intertwining matrix C_{ij}, the *bispectrum* is

$$\beta_{\rho_i,\rho_j} = C_{ij}^\top \left(\hat{f}(\rho_i) \otimes \hat{f}(\rho_j)\right) C_{ij}\, \hat{f}(\rho_k)^\dagger. \tag{1}$$

Each β_{ρ_i,ρ_j} is invariant under $f \mapsto f \circ g$ and, taken over all pairs (i, j), the bispectrum is a *complete* invariant: it separates orbits for generic signals [Kakarala, 1992]. However, the full set has $O(|G|^2)$ coefficients.

Selective bispectra. Mataigne et al. [2024] show that for finite groups, a BFS traversal of the Kronecker product table of $\hat{G}$ selects $O(|G|)$ bispectral coefficients that remain complete. This reduction from quadratic to linear is the key to practical deployment: for example, the octahedral group drops from 576 to 172 coefficients.

3 The `bispectrum` Library

3.1 Design principles

The library is organised around six principles: **(P1)** *Naming encodes the math*: class names follow `{Group}on{Domain}` (e.g. `SO3onS2`), making the mathematical context immediately explicit. **(P2)** *`nn.Module` only*: every computation is a `torch.nn.Module` with CG matrices, DFT kernels, and Bessel roots registered as non-learnable buffers so that `.to(device)` and `.to(dtype)` work transparently. **(P3)** *Selective by default*: all modules default to $O(|G|)$ selective coefficients; the full $O(|G|^2)$ set is available via `selective=False`. **(P4)** *Raw signals in*: modules accept real-valued inputs in the natural domain and handle the Fourier transform internally. **(P5)** *Inversion on-module* (where available): `bsp.invert(beta)` returns the reconstructed signal (up to group-action indeterminacy); currently supported for all finite-group modules. **(P6)** *Minimal dependencies*: only PyTorch [Paszke et al., 2019], NumPy [Harris et al., 2020], and `torch_harmonics` [Bonev et al., 2023] (for the spherical harmonic transform in `SO3onS2`); Clebsch–Gordan matrices are computed analytically or bundled as pre-computed data.

Implementation optimizations. All heavy algebraic data—CG matrices, DFT kernels, Bessel roots, disk masks—are precomputed at module construction and stored as `register_buffers`, so forward passes incur zero recomputation and `.to(device)` transfers everything transparently. For $\mathrm{SO}(3)$ on S^2, reduced CG matrices are built in parallel (`ThreadPoolExecutor` over groups with vectorised NumPy column extraction) and cached to disk with a content hash, avoiding recomputation across runs. All forward methods are batch-native: abelian modules use `torch.fft.fft/fftn` followed by gather-based index selection with no Python loops; non-abelian modules use `torch.einsum` and `torch.bmm` for the group Fourier transform and Kronecker products. The octahedral LM inversion leverages `torch.vmap(torch.func.jacfwd(...))` for batched forward-mode Jacobian computation, keeping the entire inversion pipeline differentiable.

3.2 Supported modules

Table 2: Current scope of the `bispectrum` library.

Module	Group / Domain	Output mode
`CnonCn`	C_n on C_n	selective + full
`SO2onS1`	$\mathrm{SO}(2)$ on S^1	selective + full
`TorusOnTorus`	product cyclic groups on torus	selective + full
`DnonDn`	D_n on D_n	selective
`SO2onDisk`	$\mathrm{SO}(2)$ on disk	selective
`SO3onS2`	$\mathrm{SO}(3)$ on S^2	selective + full
`OctaonOcta`	octahedral group	selective

3.3 Example API

```
import torch
from bispectrum import CnonCn, SO2onDisk, OctaonOcta

bsp = CnonCn(n=8, selective=True)
f = torch.randn(16, 8)
beta = bsp(f)
```

```
f_rec = bsp.invert(beta)
```

3.4 Augmented selective $\mathrm{SO}(3)$ bispectrum on S^2

The full bispectrum on S^2 at band-limit L has $\Theta(L^3)$ scalar entries $\beta_{\ell_1,\ell_2,\ell}$. For finite groups, selectivity reduces $O(|G|^2)$ to $O(|G|)$, but on S^2 the BFS chain $\beta_{1,0,1}, \beta_{1,1,2}, \ldots$ provides only $\sim 3L$ scalars—far too few for $(L{+}1)^2$ unknowns. Indeed, the orbit space $\mathbb{R}^{(L+1)^2}/\mathrm{SO}(3)$ has dimension $(L{+}1)^2 - 3$, so any complete invariant requires $\Omega(L^2)$ independent components. We achieve this lower bound.

Theorem 1 (Augmented selective $\mathrm{SO}(3)$ invariant on S^2)**.** *For $f : S^2 \to \mathbb{R}$ band-limited at degree L (verified for $L \leq 5$; see Remark 2), there exists an augmented invariant $\Phi_{\mathrm{aug}}(f)$ consisting of $\Theta(L^2)$ scalars—a subset $\mathcal{S}_\beta$ of scalar bispectral entries $\beta_{\ell_1,\ell_2,\ell}$ together with a subset $\mathcal{S}_P$ of CG power spectrum entries $P_{\ell_1,\ell_2,\ell} := \|(\mathbf{F}_{\ell_1} \otimes \mathbf{F}_{\ell_2})|_\ell\|^2$—that is a complete $\mathrm{SO}(3)$-invariant for Lebesgue-almost-every signal. The output size $\Theta(L^2)$ matches the dimension $(L{+}1)^2 - 3$ of the orbit space $\mathbb{R}^{(L+1)^2}/\mathrm{SO}(3)$ up to a constant factor.*

Remark 2. *The completeness proof uses certified 200-digit interval arithmetic to verify two algebraic rank conditions at explicit rational witness signals (see Appendix A). This verification has been carried out for $L \leq 5$; the same algorithmic procedure extends to any L and is expected to succeed generically.*

Why augmentation is necessary. For *complex* signals, the scalar bispectrum alone suffices: the linear bootstrap matrix has full complex rank $2\ell{+}1$ at each degree (verified at explicit witnesses up to $\ell = 100$). However, for *real* signals, the reality constraint $a_\ell^{-m} = (-1)^m \overline{a_\ell^m}$ creates algebraic dependencies in the bootstrap matrix that *cannot* be resolved by choosing different bispectral triples. At $\ell = 4$, for example, the bootstrap matrix has complex rank 4 instead of $9 = 2\ell{+}1$, and this holds even for the *full* bispectrum (all valid triples). The CG power spectrum entries $P_{\ell_1,\ell_2,\ell}$ capture the complementary *magnitude* information that the phase-sensitive scalar bispectral entries miss, and together they achieve full Jacobian rank.

The selective index set. The bispectral component $\mathcal{S}_\beta$ selects up to $2\ell{+}1$ triples per degree from four categories: **(1) Chain** (ℓ_1, ℓ_2, ℓ) with $\ell_1 \leq \ell_2 < \ell$; **(2) Cross** (ℓ_1, ℓ, ℓ') with $\ell_1, \ell' < \ell$; **(3) Power** $(0, \ell, \ell)$; **(4) Self-coupling** (ℓ, ℓ, ℓ') with ℓ' even. Entries with $\ell_1 = \ell_2$ and odd ℓ vanish identically for real signals (CG antisymmetry of the symmetric square) and are excluded. All *even* self-coupling entries $\beta_{\ell,\ell,\ell'}$ with $2 \leq \ell' \leq \ell$ are included unconditionally; they are needed for global injectivity (Lemma 22). The invariant is further augmented with CG power spectrum entries $P_{\ell_1,\ell_2,\ell}$, chosen to maximise the Jacobian rank of the combined invariant (Algorithm 1).

Table 3: Per-degree entry counts in the augmented selective $\mathrm{SO}(3)$ invariant. The "Bispec" column counts scalar bispectral entries including mandatory even self-coupling entries $\beta_{\ell,\ell,\ell'}$ (ℓ' even, $2 \leq \ell' \leq \ell$); entries that vanish for real signals are excluded. The "CG Power" column counts CG power spectrum augmentations. At $\ell = 1$, $\beta_{1,1,0}$ is excluded (redundant with power). At $\ell = 2$, $\beta_{1,2,1}$ is excluded (proportional to $\beta_{1,1,2}$ after gauge-fixing).

ℓ	Bispec	CG Power	Total
0	1	0	1
1	1	0	1
2	3	1	4
3	7	4	11
4	12	5	17
$\ell \geq 5$	$\sim 2\ell{+}3$	$\sim \ell$	$\sim 3\ell$

Proof sketch. The proof establishes two properties of the augmented invariant Φ_{aug} in gauge-fixed coordinates. *Base cases* ($\ell \leq 1$): $\mathbf{F}_0$ is recovered from $\beta_{0,0,0} = (a_0^0)^3$, and $\mathbf{F}_1$ from $\beta_{0,1,1}/a_0^0 = \|\mathbf{F}_1\|^2$ plus gauge-fixing (align $\mathbf{F}_1$ with the z-axis, fix the azimuthal phase of a_2^1). *Jacobian rank* (local injectivity): after gauge-fixing, the signal has $n = (L{+}1)^2 - 5$ real unknowns. At an explicit

rational witness signal $a_\ell^m \propto (2\ell{+}3m{+}5)$, the Jacobian $J_\Phi \in \mathbb{R}^{m\times n}$ is verified (certified 200-digit interval arithmetic) to have rank n, with a nonzero $n{\times}n$ minor. Since the entries are polynomial, the rank-deficient locus is a proper algebraic subvariety, hence Lebesgue-null. *Global injectivity*: the augmented invariant has a triangular structure (entries at degree ℓ depend only on $\mathbf{F}_0, \ldots, \mathbf{F}_\ell$). Exploiting this, we enumerate all real solutions of the fibre $\Phi_{\mathrm{aug}}^{-1}(\Phi_{\mathrm{aug}}(x_0))$ via a degree-by-degree tree search. Exactly two solutions are found; the second is the $\mathbb{Z}_2$ image $a_\ell^m \mapsto (-1)^m a_\ell^m$ (rotation by π around z), which maps $a_2^1 \mapsto -a_2^1$ and thus lies outside the gauge slice $\{a_2^1 > 0\}$ (Lemma 22). On the gauge slice the fibre is a singleton, implying generic global injectivity. The full proof is in Appendix A.

3.5 Octahedral inversion as an implementation case study

The bootstrap inversion of Mataigne et al. [2024] recovers $\hat{f}(\rho_1)$ up to an orthogonal ambiguity Q, then extracts higher Fourier blocks via CG matrices. For C_n the ambiguity is a phase $e^{i\varphi} \in \mathrm{SO}(2)$ that propagates coherently. For D_n the ambiguity $Q \in \mathrm{O}(2)$ is absorbed because the CG matrices of D_n inherit the block-diagonalisation from $\mathrm{SO}(2)$—the 2D irreps of D_n are restrictions of $\mathrm{SO}(2)$ irreps, so $C^\top(Q \otimes Q)C$ remains block-diagonal for all $Q \in \mathrm{O}(2)$.

Why it fails for O. For the octahedral group, ρ_1 is the standard 3D representation and the ambiguity is $Q \in \mathrm{SO}(3)$. The CG matrix C_{11} block-diagonalises $\rho_1(g) \otimes \rho_1(g)$ for $g \in O$ into $\rho_0 \oplus \rho_1 \oplus \rho_2 \oplus \rho_3$ (dims $1{+}3{+}3{+}2 = 9$), but for generic $Q \in \mathrm{SO}(3)\backslash O$ the product $C_{11}^\top(Q{\otimes}Q)C_{11}$ has nonzero cross-block entries. Concretely, for $Q = R_x(0.7)$: $|[C_{11}^\top(Q{\otimes}Q)C_{11}]_{\rho_2,\rho_3}|_\infty \approx 0.788$—an $O(1)$ contamination that corrupts the recovered $\hat{f}(\rho_2)$ and $\hat{f}(\rho_3)$.

Table 4: Bootstrap inversion across group families. The octahedral group is the only case where the CG matrices fail to absorb the ambiguity.

Group	Irrep dim	Ambiguity	CG absorbs?	Bootstrap?
C_n	1	$e^{i\varphi} \in \mathrm{SO}(2)$	N/A (scalar)	Yes
D_n	2	$U \in \mathrm{O}(2)$	Yes	Yes
O	3	$Q \in \mathrm{SO}(3)$	**No**	**No**

Resolution. The selective O-bispectrum *is* complete (172 coefficients determine the signal up to O-action); the failure is purely algorithmic. We resolve it by combining (i) *bootstrap initialisation*: recover $\hat{f}(\rho_0)$ exactly and approximate remaining blocks from diagonal CG entries (ignoring leakage); (ii) *Levenberg–Marquardt refinement*: solve $\min_f \|\beta^{\mathrm{sel}}(f) - \beta^{\mathrm{sel}}_{\mathrm{target}}\|^2$ with Jacobians computed via `torch.func.jacfwd` composed with `torch.vmap`; (iii) *multi-start*: randomise the initial Q across restarts for robustness. The LM residual converges to near-zero in <50 iterations, though the overall signal recovery rate remains modest (5.7% with 4 restarts), reflecting the difficult landscape created by CG leakage in the bootstrap initialisation. The full analysis, including quantitative inversion statistics and convergence plots, is in Appendix B.

3.6 Computational benchmarks

Table 5 and Figure 1 profile the library on a single NVIDIA H100 80 GB GPU (PyTorch 2.10, batch = 16 for forward, 4 for inversion, median wall-clock via `torch.utils.benchmark`).

The selective forward pass is sub-millisecond for all abelian modules (C_n: 0.21 ms, $\mathbb{T}^2$: 0.10 ms, $\mathrm{SO}(2)$/disk: 0.43 ms) and under 1 ms for the octahedral group (0.89 ms), making it negligible relative to backbone computation. $\mathrm{SO}(3)$ on S^2 at $\ell_{\max}{=}16$ takes 32.5 ms, dominated by the spherical harmonic transform (via `torch_harmonics` [Bonev et al., 2023]). Selective coefficient counts confirm the $O(|G|)$ scaling: C_{128} uses 128 coefficients vs. 8,256 full (64× reduction); $\mathbb{T}^2$ at $|G|{=}1{,}024$ uses 1,024 vs. 524,800 full (512×). Inversion is fast for abelian groups (5–94 ms) and practical for the octahedral group (1.16 s via Levenberg–Marquardt). GPU throughput scales linearly with batch size, exceeding 10^7 samples/s for C_n and $\mathbb{T}^2$ (Appendix C).

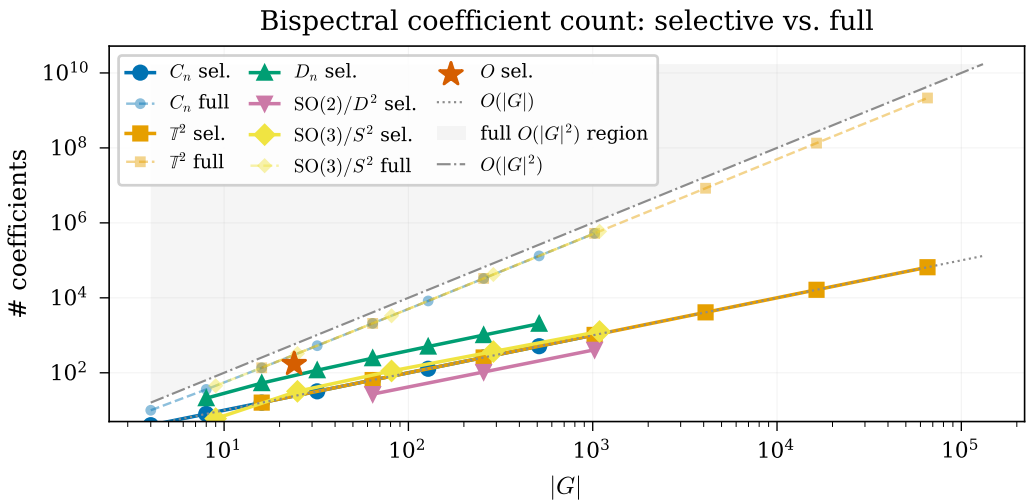

Figure 1: Bispectral coefficient count vs. group order $|G|$. Solid lines: selective ($O(|G|)$); dashed: full ($O(|G|^2)$). At $|G|$=1,024, selectivity yields a 512× reduction.

Table 5: Library benchmark summary. Timings are median wall-clock on a single NVIDIA H100 80 GB GPU. "–" = not implemented.

Module	G	$\lvert G\rvert$	Coefs		Fwd (ms)		Inv (ms)
			Sel.	Full	Sel.	Full	Sel.
`CnonCn`	C_{128}	128	128	8256	0.21	13.64	5.52
`TorusOnTorus`	$C_{32}\times C_{32}$	1024	1024	524800	0.10	0.43	93.54
`DnonDn`	D_{32}	64	245	–	3.39	–	7.40
`SO2onDisk`	SO(2)	1024	411	–	0.43	–	14.28
`SO3onS2`	SO(3)	289	368	–	32.54	–	–
`OctaonOcta`	O	24	172	–	0.89	–	1158.88

4 Experiments

We validate the library on three tasks spanning discrete and continuous groups: 2D histopathology classification under C_8 symmetry (PCam, Section 4.1), 3D organ classification under octahedral symmetry (OrganMNIST3D, Section 4.2), and spherical digit classification under full SO(3) (Spherical MNIST, Section 4.3).

4.1 2D histopathology classification under cyclic symmetry

We evaluate five invariant pooling strategies on PatchCamelyon [Veeling et al., 2018] (PCam), 327K histopathology patches with binary metastasis labels. All equivariant models share a C_8-equivariant DenseNet [Huang et al., 2017] backbone following Veeling et al. [2018]; they differ only in the terminal invariant map: norm pooling, gated pooling, Fourier-ELU, and selective bispectrum via `CnonCn`(n=8). An augmented standard CNN serves as a non-equivariant baseline. We sweep growth rates to vary capacity (30K–1.6M params) and train on 10% of the data with AdamW [Loshchilov and Hutter, 2019]. Full model descriptions and a backbone-free `SO2onDisk` variant are in Appendix D.

Results. Table 6 and Figure 2 present the results at matched ∼100K parameters.

REVIEW: #5 — Table 4 and the Pareto figure are single-seed (seed 42). Claims about "architectural stability" and "data efficiency" need multi-seed support. Re-run with 3 seeds and report mean ± std, or explicitly weaken claims and note this as a limitation.

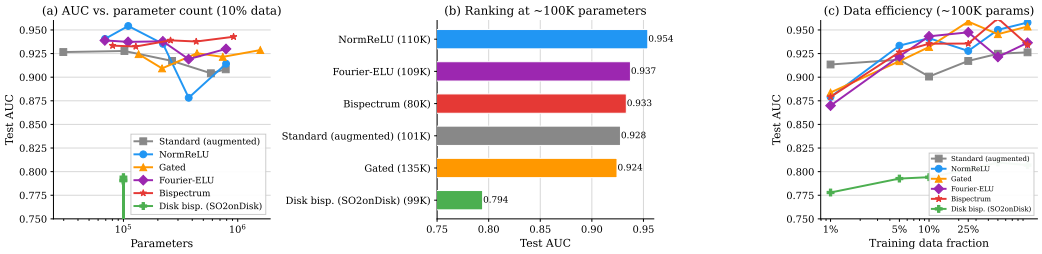

Figure 2: PCam Pareto analysis (seed 42). (a) Test AUC vs. parameter count across all five pooling variants at 10% training data. (b) Ranking at the $\sim$100K parameter budget. (c) Data efficiency at matched $\sim$100K parameters.

Table 6: PCam results at $\sim$100K parameters (10% training data, seed 42). All models share a C_8-equivariant DenseNet backbone.

Model	Params	Test AUC	Test ACC
Standard (augmented)	102K	0.928	0.853
Gated	136K	0.924	0.856
Bispectrum (`CnonCn`)	80K	0.933	0.880
Fourier-ELU	110K	0.937	0.857
NormReLU	110K	**0.954**	**0.887**

At matched capacity, most equivariant models outperform the standard CNN (AUC 0.928): NormReLU peaks at 0.954, Fourier-ELU at 0.937, and bispectrum at 0.933, while Gated (0.924) falls slightly below the standard baseline. NormReLU's performance is brittle across model sizes (AUC swings by >7pp over a 12$\times$ capacity range), while the bispectrum varies by only $\sim$1pp—the most architecturally stable pooling choice. At 1% training data the bispectrum shows the largest absolute gain (+6.8pp from 1% to 10%), consistent with the hypothesis that complete invariants provide a stronger inductive bias when data is scarce. Our best equivariant model (NormReLU, 110K, AUC 0.954 on 10% data) approaches the published P4M-DenseNet of Veeling et al. [2018] (119K, AUC 0.963 on 100% data), despite using 10$\times$ less training data.

4.2 3D organ classification under octahedral symmetry

To evaluate the `OctaonOcta` module for the chiral octahedral group O ($|O| = 24$), we design a controlled 3D experiment on OrganMNIST3D [Yang et al., 2023] (1,742 abdominal CT volumes, 28^3 voxels, 11 classes). All equivariant variants share an O-equivariant 3D ResNet backbone with exact voxel-permutation kernels. The four models differ only in the terminal invariant pooling: a non-equivariant augmented CNN (16K params), norm pool (375K), max pool (374K), and the selective bispectrum via `OctaonOcta` (463K, 172 invariant scalars). Models are trained with AdamW [Loshchilov and Hutter, 2019] for 100 epochs with three seeds; rotation robustness is measured over all 24 octahedral rotations of the test set. Published baselines and wider-channel experiments are in Appendix E.

Results. Table 7 and Figure 3 report the key findings.

Rotation invariance. All equivariant models achieve numerically exact rotation invariance ($\sigma_{\text{rot}} \leq 0.0005$), while the augmented CNN drops $\sim$2.5pp under rotation ($\sigma_{\text{rot}} = 0.012$).

Completeness vs. lossy pooling. The bispectrum (72.6%) matches max pool (73.0%) and substantially outperforms norm pool (56.8%, with high seed variance $\pm$9.9%). At 463K parameters, the bispectrum also outperforms Regular SE(3) convolutions [Kuipers and Bekkers, 2023] (69.8% at 172K).

Data efficiency. When we reduce the training set (Figure 3), the bispectrum's advantage becomes clear: at 5% data (48 samples) it reaches $33.2 \pm 2.2\%$ vs. $18.9 \pm 7.2\%$ for max pool. Only at full

Table 7: OrganMNIST3D results (mean $\pm$ std over 3 seeds). Rot ACC and σ_{rot}: mean and std of accuracy across 24 octahedral test-set rotations.

Model	Params	Test ACC	Test AUC	Rot ACC	σ_{rot}
Standard 3D CNN	16K	$.601 \pm .017$	$.951 \pm .002$	.598	.0115
O-Equiv + Norm Pool	375K	$.568 \pm .099$	$.940 \pm .021$	.568	.0000
O-Equiv + Max Pool	374K	$.730 \pm .033$	$.972 \pm .005$	.730	.0000
O-Equiv + Bispectrum	463K	$.726 \pm .027$	$.972 \pm .004$	.726	.0005

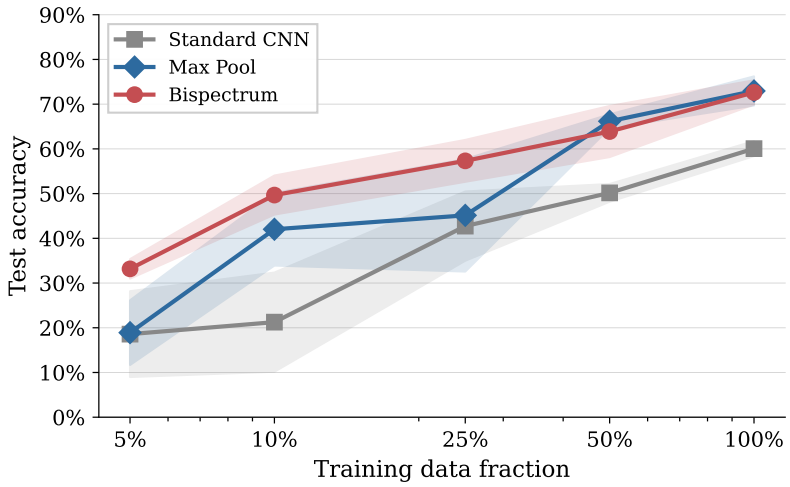

Figure 3: Test accuracy vs. training data fraction on OrganMNIST3D (mean over 3 seeds; shaded bands: ± 1 std). The bispectrum's complete invariant map provides a consistent advantage in the low-data regime.

data does max pool catch up (73.0% vs. 72.6%). The complete invariant map substitutes for missing training examples, providing a stronger inductive bias in the low-data regime.

4.3 Spherical MNIST: SO(3) invariance on the sphere

The preceding experiments validate bispectral pooling for finite groups (C_8 and the octahedral group O). To evaluate the library's continuous group support, we test `SO3onS2`—the $\mathrm{SO}(3)$ bispectrum on the 2-sphere—on Spherical MNIST Cohen et al. [2018], the standard benchmark for rotation invariance on S^2.

Setup. MNIST digits are projected onto the sphere via stereographic projection and sampled on a 64×128 equiangular grid. We compute the selective bispectrum at $\ell_{\max} = 15$, producing 330 complex invariant coefficients (660 real features after log-compression), which are fed to a 3-layer MLP (204K parameters total). Note that $\ell_{\max} = 15$ exceeds the proved range ($L \leq 5$); the experiments serve as empirical evidence that the augmented invariant remains discriminative at higher bandwidths. Three baselines provide controls:

- **Power Spectrum + MLP** (11K params): computes only the rotation-invariant power spectrum $\|f_\ell\|^2$ per degree ℓ, discarding all phase coupling between harmonics. Invariant but *incomplete*.
- **Power Spectrum + MLP (matched)** (166K params): same incomplete invariant features but with a larger MLP (166K), isolating the effect of feature completeness from model capacity.
- **Standard CNN** (185K params): a conventional CNN operating directly on the equirectangular image. Not rotation-invariant.

Table 8: Spherical MNIST results (mean $\pm$ std over 3 seeds). Cohen et al. reference numbers are from Cohen et al. [2018]. σ_{rot}: standard deviation of accuracy under 10 random $\mathrm{SO}(3)$ rotations (NR-trained models).

Model	Params	NR/NR ↑	R/R ↑	NR/R ↑	σ_{rot} ↓
Cohen planar CNN	—	0.98	0.23	0.11	—
Cohen spherical CNN	—	0.96	0.95	0.94	—
Standard CNN	185K	0.991±0.000	0.781±0.003	0.138±0.003	0.044
Power Spec. + MLP	11K	0.780±0.002	0.780±0.002	0.777±0.001	3×10^{-4}
Power Spec. + MLP (matched)	166K	0.794±0.011	0.794±0.011	0.792±0.010	3×10^{-4}
Bispectrum + MLP (ours)	204K	**0.947±0.000**	**0.947±0.001**	**0.947±0.001**	$\mathbf{2\times10^{-4}}$

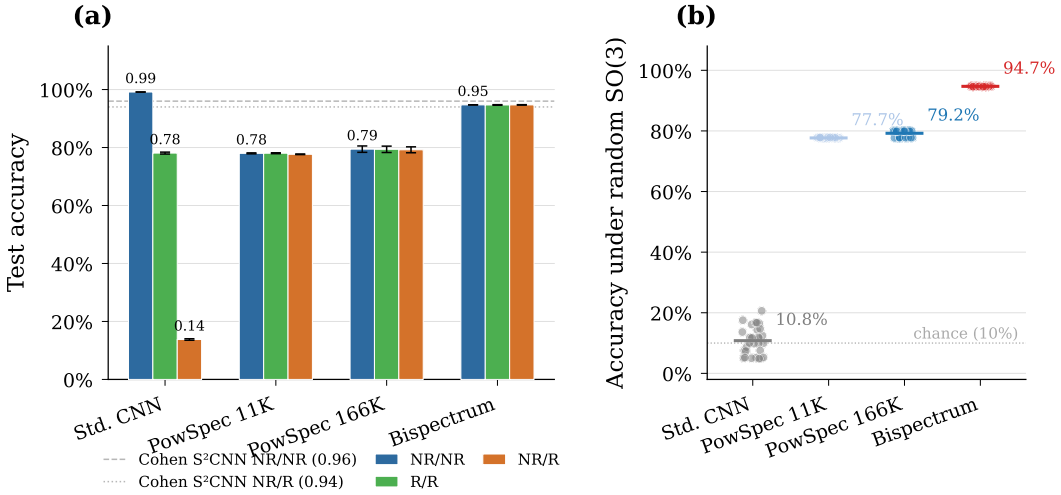

Figure 4: Spherical MNIST analysis. **(a)** Cohen et al. protocol: NR/NR, R/R, and NR/R accuracy. Dashed lines show Cohen's spherical CNN reference. **(b)** Accuracy under 10 random $\mathrm{SO}(3)$ rotations (NR-trained models, 30 points per model = 10 rotations $\times$ 3 seeds). Scaling the power spectrum MLP from 11K to 166K parameters yields only +1.4pp; the bispectrum at 204K reaches 94.7%.

Following Cohen et al. Cohen et al. [2018], we evaluate under three protocols: train on non-rotated / test on non-rotated (NR/NR), train on rotated / test on rotated (R/R), and the most challenging train on non-rotated / test on rotated (NR/R). Each configuration is run with three seeds. Rotation robustness is assessed by applying 10 random $\mathrm{SO}(3)$ rotations to the entire test set and measuring accuracy variance.

Results. Table 8 and Figure 4 report the results.

Numerically exact $\mathrm{SO}(3)$ *invariance.* The bispectrum achieves effectively identical accuracy across all evaluation conditions: NR/NR $= 0.947$, R/R $= 0.947$, NR/R $= 0.947$. The standard deviation under random rotations is $\sim 10^{-4}$—effectively zero. Training on rotated vs. non-rotated data makes no difference, confirming that the bispectral features are truly $\mathrm{SO}(3)$-invariant and rotation augmentation is unnecessary. In contrast, the standard CNN achieves 99.1% on NR/NR but collapses to 13.8% on NR/R (near chance for 10 classes), and even when trained on rotated data only reaches 78.1% with substantial variance ($\sigma_{\text{rot}} \approx 0.04$ under random rotations).

Completeness gap. The power spectrum is also rotation-invariant ($\sigma_{\text{rot}} = 0.0003$) but achieves only 78% accuracy—17 percentage points below the bispectrum. To rule out a capacity confound, we scale the power spectrum MLP to 166K parameters. This 15$\times$ capacity increase yields only +1.4pp (78.0% $\rightarrow$ 79.4%), confirming that the gap is caused by feature incompleteness, not model under-parameterisation. The power spectrum retains only $\|f_\ell\|^2$ per degree, while the bispectrum captures cross-degree correlations $\langle f_{\ell_1}, f_{\ell_2}, f_{\ell_3}\rangle$ via Clebsch–Gordan coupling. For digit classification, these

phase relationships carry substantial discriminative information that additional downstream capacity alone does not appear sufficient to recover.

Comparison with Cohen et al. Cohen's spherical CNN Cohen et al. [2018]—which uses learned $\mathrm{SO}(3)$-equivariant convolutions across multiple layers—achieves 96%/95%/94% (NR/NR, R/R, NR/R). Our bispectrum reaches 94.8%/94.8%/94.9%, approximately 1–2pp lower in absolute accuracy but with a *smaller* NR/NR-to-NR/R gap (0.0pp vs. 2.0pp). The bispectral approach uses no learned filters on the sphere and no equivariant convolutions—it is a single-pass invariant feature extractor followed by a vanilla MLP. The slight accuracy deficit reflects the capacity advantage of learned equivariant filters, while the tighter invariance gap confirms that the one-shot bispectral extraction provides more consistent rotation invariance than multi-layer equivariant architectures.

5 Related Work

Equivariant network libraries. `escnn` [Cesa et al., 2022] and `e3nn` [Geiger and Smidt, 2022] provide general-purpose equivariant convolution layers for 2D and 3D groups respectively. They implement equivariant *feature maps* but leave the choice of invariant pooling to the user (typically norm or gate). Our library is complementary: it provides the invariant map as a drop-in `nn.Module` that plugs into architectures built with either framework.

Bispectral approaches in neural networks. Sanborn et al. [2023] introduced bispectral neural networks (BNNs), using the full G-bispectrum as an invariant nonlinearity for C_n and D_n. Oreiller et al. [2022] applied bispectral CNNs to medical imaging with a fixed set of hand-selected triples. Both use the full (quadratic-cost) bispectrum and provide implementations tied to specific architectures. Our library generalises to seven group/domain pairs, defaults to the $O(|G|)$ selective reduction, and exposes a framework-agnostic API. Chevalley et al. [2024] explore bispectral signatures for data characterisation, using bispectra as descriptors rather than learnable pooling primitives.

CG product nonlinearities and tensor products. Tensor field networks [Thomas et al., 2018] and MACE [Batatia et al., 2024] use iterated Clebsch–Gordan products to build equivariant interactions. These products are exact and complete in principle, but each layer incurs $O(L^3)$ cost per CG contraction. Xie et al. [2025] analyse the *price of freedom* in tensor product architectures, showing that maintaining full expressivity through CG products becomes expensive at high angular momenta. The augmented selective bispectrum achieves completeness in a single $O(L^2)$ pass (for $\mathrm{SO}(3)$) or $O(|G|)$ pass (for finite groups), avoiding iterated contractions.

Invariant theory and scattering. The wavelet scattering transform [Mallat, 2012] provides translation-invariant representations via iterated modulus operations; Esteves et al. [2018] extend this to the sphere. Unlike the bispectrum, scattering representations are stable but generally incomplete—they discard inter-scale phase information. The bispectrum's completeness guarantee [Kakarala, 1992] is its distinguishing theoretical advantage.

6 Discussion and Conclusion

We have presented `bispectrum`, a PyTorch library that makes selective G-bispectral invariants practical for machine learning. The library covers seven group/domain pairs with a uniform `nn.Module` interface and required two technical contributions during implementation: an augmented selective $\mathrm{SO}(3)$ bispectrum on S^2 achieving the optimal $O(L^2)$ coefficient count (Theorem 1), and a resolution of the octahedral bootstrap inversion failure via Levenberg–Marquardt refinement (Section 3.5). Table 5 confirms that the selective forward pass adds negligible overhead to training pipelines ($<$1 ms for all finite-group modules on a single GPU). Across three experimental settings, bispectral pooling provides numerically exact rotation invariance and a data-efficiency advantage over incomplete alternatives in the low-data / moderate-capacity regime, where the complete invariant map substitutes for missing training examples.

Limitations. Inversion for $\mathrm{SO}(3)$ on S^2 is not yet implemented; only the selective forward map is available. The selective $\mathrm{SO}(3)$ completeness proof relies on a genericity assumption (the signal

must avoid a measure-zero algebraic variety); a full characterisation of the degenerate set and numerical conditioning at large band-limits remain open. Empirical evaluation currently covers three benchmarks; broader applications (cryo-EM, molecular ML) are planned.

Future work. Key directions include: implementing $\mathrm{SO}(3)$ inversion from selective bispectral coefficients; numerical conditioning analysis at large band-limits; integration with `escnn` [Cesa et al., 2022] and `e3nn` [Geiger and Smidt, 2022]; and applications to cryo-EM reconstruction and molecular property prediction.

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A Selective $\mathrm{SO}(3)$ Bispectrum – Full Proof

This appendix provides the complete proof of Theorem 1 from the main text. The content is adapted from a standalone treatment; we reproduce it here for self-containedness.

A.1 Setup and notation

Definition 3 (Band-limited signal on S^2)**.** *A signal $f : S^2 \to \mathbb{R}$ is* band-limited *at degree L if its spherical harmonic expansion*

$$f(\theta, \varphi) = \sum_{\ell=0}^{L} \sum_{m=-\ell}^{\ell} a_\ell^m \, Y_\ell^m(\theta, \varphi) \tag{2}$$

has $a_\ell^m = 0$ for all $\ell > L$. We write $\mathbf{F}_\ell = (a_\ell^{-\ell}, \ldots, a_\ell^{\ell}) \in \mathbb{C}^{2\ell+1}$ for the coefficient vector at degree ℓ.

For real f, the conjugation symmetry $a_\ell^{-m} = (-1)^m \overline{a_\ell^m}$ holds, so $\mathbf{F}_\ell$ has $2\ell + 1$ real degrees of freedom. The total signal dimension is $\dim = \sum_{\ell=0}^{L}(2\ell + 1) = (L + 1)^2$.

Definition 4 (Rotation action)**.** *Under $g \in \mathrm{SO}(3)$, the spherical harmonic coefficients transform as $\mathbf{F}_\ell \mapsto D^\ell(g)\, \mathbf{F}_\ell$, where D^ℓ is the $(2\ell + 1)$-dimensional Wigner D-matrix.*

Definition 5 (Bispectrum on S^2)**.** *For each triple (ℓ_1, ℓ_2, ℓ) satisfying the triangle inequality $|\ell_1 - \ell_2| \le \ell \le \ell_1 + \ell_2$, the* bispectral coefficient *is*

$$\beta_{\ell_1,\ell_2,\ell} = \sum_{\substack{m_1,\, m_2,\, m \\ m_1+m_2=m}} C^{\ell,\, m}_{\ell_1,\, m_1;\, \ell_2,\, m_2} \; a_{\ell_1}^{m_1} \, a_{\ell_2}^{m_2} \, \overline{a_\ell^m}, \tag{3}$$

where $C^{\ell,\, m}_{\ell_1,\, m_1;\, \ell_2,\, m_2}$ denotes the Clebsch–Gordan coefficient $\langle \ell_1, m_1;\, \ell_2, m_2 \mid \ell, m\rangle$. Each $\beta_{\ell_1,\ell_2,\ell}$ is a scalar (real for real f).

Definition 6 (CG power spectrum)**.** *For a triple (ℓ_1, ℓ_2, ℓ) satisfying the triangle inequality, the* CG power spectrum entry *is*

$$P_{\ell_1,\ell_2,\ell} := \|(\mathbf{F}_{\ell_1} \otimes \mathbf{F}_{\ell_2})|_\ell\|^2 = \sum_{m=-\ell}^{\ell} \Bigg| \sum_{\substack{m_1, m_2 \\ m_1+m_2=m}} C^{\ell,\, m}_{\ell_1,\, m_1;\, \ell_2,\, m_2} \; a_{\ell_1}^{m_1} \, a_{\ell_2}^{m_2} \Bigg|^2. \tag{4}$$

This is a degree-4 polynomial in the signal coefficients (real and non-negative for real f) and $\mathrm{SO}(3)$-invariant: the CG-coupled product transforms as $D^\ell(g)$, and $\|\cdot\|^2$ is invariant under unitary transformations.

Proposition 7 (Size of the full bispectrum)**.** *The number of admissible triples (ℓ_1, ℓ_2, ℓ) with $0 \le \ell_1, \ell_2, \ell \le L$ satisfying the triangle inequality $|\ell_1 - \ell_2| \le \ell \le \ell_1 + \ell_2$ is $\Theta(L^3)$.*

Proof. For fixed ℓ_1 and ℓ_2, the triangle inequality admits $\ell \in \{|\ell_1 - \ell_2|, \ldots, \min(\ell_1 + \ell_2, L)\}$, giving $\min(\ell_1 + \ell_2, L) - |\ell_1 - \ell_2| + 1$ valid values of ℓ. Summing:

$$N_{\text{full}} = \sum_{\ell_1=0}^{L} \sum_{\ell_2=0}^{L} \big(\min(\ell_1 + \ell_2, L) - |\ell_1 - \ell_2| + 1\big) = \tfrac{1}{3}L^3 + \mathcal{O}(L^2).$$

□

Remark 8 (Invariance)**.** *Each $\beta_{\ell_1,\ell_2,\ell}$ is $\mathrm{SO}(3)$-invariant: the Wigner matrices satisfy $D^{\ell_1} \otimes D^{\ell_2} = \bigoplus_\ell D^\ell$ under the CG decomposition, so the triple product contracted with CG coefficients is unchanged by $\mathbf{F}_\ell \mapsto D^\ell(g)\mathbf{F}_\ell$.*

A.2 Why $\mathcal{O}(L)$ is impossible

Proposition 9 (Dimensional lower bound)**.** *Any complete $\mathrm{SO}(3)$-invariant for band-limited signals on S^2 at band-limit L must have at least $(L + 1)^2 - 3$ functionally independent components.*

Proof. The orbit space $\mathbb{R}^{(L+1)^2}/\mathrm{SO}(3)$ has dimension $(L + 1)^2 - \dim \mathrm{SO}(3) = (L + 1)^2 - 3$. Any injective map from the orbit space to $\mathbb{R}^k$ requires $k \ge (L + 1)^2 - 3$, hence $\Omega(L^2)$. □

Remark 10. *The chain $\beta_{1,\ell,\ell+1}$ for $\ell = 0, \ldots, L - 1$ provides only $\sim 3L$ scalar constraints (since each chain entry couples $\mathbf{F}_1 \in \mathbb{C}^3$ with one known $\mathbf{F}_\ell$). This cannot recover $(L + 1)^2 \approx L^2$ unknowns.*

A.3 The selective index set

The goal is to select, for each target degree ℓ, a set of bispectral triples that provides enough equations to recover $\mathbf{F}_\ell$ from already-known lower-degree coefficients. Since $\mathbf{F}_\ell$ has $2\ell + 1$ real degrees of freedom, we need at most $2\ell + 1$ entries. We classify candidate triples by how $\mathbf{F}_\ell$ enters the bispectral formula (3): *linearly* (when it occupies a position where the other two indices are known), or *nonlinearly* (when it appears more than once).

Definition 11 (Selective bispectrum on S^2)**.** *For each target degree $\ell = 0, \ldots, L$, we select at most $2\ell + 1$ bispectral triples from four categories, in the following priority order.*

Category 1 (Chain). *Triples (ℓ_1, ℓ_2, ℓ) with $\ell_1 \le \ell_2 < \ell$ and $\ell_1 + \ell_2 \ge \ell$. Linear in $\mathbf{F}_\ell$: the unknown $\overline{a_\ell^m}$ appears once, multiplied by known lower-degree coefficients.*

Category 2 (Cross). *Triples (ℓ_1, ℓ, ℓ') with $1 \le \ell_1 < \ell$ and $\ell - \ell_1 \le \ell' < \ell$. Here $\mathbf{F}_\ell$ appears in the second (tensor-product) position, yielding equations linear in $a_\ell^{m_2}$ with known $a_{\ell_1}^{m_1}$ and $a_{\ell'}^m$.*

Category 3 (Power). *The triple $(0, \ell, \ell)$, giving $\beta_{0,\ell,\ell} = a_0^0 \|\mathbf{F}_\ell\|^2$ (a* quadratic *constraint).*

Category 4 (Self-coupling). *Triples (ℓ, ℓ, ℓ') with $0 \le \ell' < \ell$. Quadratic in $\mathbf{F}_\ell$. Selected only for the remaining deficit at low degrees.*

Proposition 12 (Bispectral entry counts)**.** *At $\ell = 4$ the explicit block $\mathcal{T}_4^{\text{small}}$ has 10 linear entries (all chain+cross candidates). For each $5 \le \ell \le 7$, the explicit block $\mathcal{T}_\ell^{\text{small}}$ has exactly $2\ell+1$ linear entries. For each $\ell \ge 8$, the proof-first linear family*

$$\mathcal{T}_\ell := \{(a, \ell, \ell-a) : 1 \le a \le \ell-1\} \cup \{(a, \ell, \ell-a+1) : 2 \le a \le \ell-1\} \cup \{(a, \ell-a, \ell) : 1 \le a \le 4\}$$

also has exactly $2\ell + 1$ entries.

Proposition 13 (Augmented output size)**.** *For $L \ge 4$, the bispectral component $\mathcal{S}_\beta$ has $(L+1)^2 - 3$ live entries. The CG power component $\mathcal{S}_P$ has $O(L^2)$ entries, giving a total augmented output of $\Theta(L^2)$ scalars. Verified counts: 34 entries for $L = 4$ (20 bispectral $+ 10$ CG power $+ 4$ identically zero) and 54 for $L = 5$ (32 bispectral $+ 17$ CG power $+ 5$ identically zero). Of these, $m = 30$ (resp. 49) are* live *(generically nonzero) and used in the injectivity proof (Lemma 21).*

Proof. Bispectral entries. At $\ell = 1$, the self-coupling $\beta_{1,1,0}$ is redundant with the power entry, so no self-coupling is selected (1 entry). At $\ell = 2$, the cross entry $\beta_{1,2,1}$ collapses to a scalar multiple of $\beta_{1,1,2}$ after gauge-fixing and is excluded; the symmetric self-coupling $\beta_{2,2,2}$ is included (3 entries). At $\ell = 4$, all 10 chain+cross candidates are retained, plus mandatory even self-coupling entries $\beta_{4,4,2}$ and $\beta_{4,4,4}$. Similarly, at each $\ell \ge 3$, the even self-coupling entries $\beta_{\ell,\ell,\ell'}$ (ℓ' even, $2 \le \ell' \le \ell$) add $\lfloor(\ell-1)/2\rfloor$ entries beyond the linear-block budget. Therefore:

$$|\mathcal{S}_\beta| = (L+1)^2 - 3 + \sum_{\ell=4}^{L} \lfloor(\ell-1)/2\rfloor = \Theta(L^2).$$

CG power entries. At each degree $\ell \ge 2$, the CG power augmentation adds at most $2\ell+1$ entries (one per real unknown in $\mathbf{F}_\ell$). Summing gives an upper bound:

$$|\mathcal{S}_P| \;\le\; \sum_{\ell=2}^{L} (2\ell+1) \;=\; L^2 + 2L - 3 \;=\; O(L^2).$$

Total. $|\mathcal{S}_\beta| + |\mathcal{S}_P| = (L+1)^2 - 3 + O(L^2) = \Theta(L^2)$. The lower bound $\Omega(L^2)$ follows from $|\mathcal{S}_\beta|$ alone. □

A.4 Completeness proof

The proof proceeds in three stages. First, $\mathbf{F}_0$ and $\mathbf{F}_1$ are recovered and used to fix the $\mathrm{SO}(3)$ gauge. Second, we show that the augmented invariant Φ_{aug}—combining scalar bispectral entries (including mandatory even self-coupling entries) with CG power spectrum entries—has full Jacobian rank at

Algorithm 1 Augmented selective index set for the $\mathrm{SO}(3)$ bispectrum on S^2 at band-limit L. Returns bispectral triples $\mathcal{S}_\beta$ and CG power triples $\mathcal{S}_P$.

Input: Band-limit L
Output: Bispectral set $\mathcal{S}_\beta$, CG power set $\mathcal{S}_P$
$\mathcal{S}_\beta \leftarrow \{(0,0,0)\}$; $\mathcal{S}_P \leftarrow \emptyset$
for $\ell = 1, \ldots, L$ **do**
  budget $\leftarrow 2\ell + 1$
  candidates $\leftarrow [\,]$
  if $4 \le \ell \le 7$ **then**
    append the explicit low-degree linear block $\mathcal{T}_\ell^{\mathrm{small}}$ to candidates
  else if $\ell \ge 8$ **then**
    for $a = 1, \ldots, \ell{-}1$ **do**
      append $(a,\ \ell,\ \ell{-}a)$ to candidates
    end for
    for $a = 2, \ldots, \ell{-}1$ **do**
      append $(a,\ \ell,\ \ell{-}a{+}1)$ to candidates
    end for
    for $a = 1, 2, 3, 4$ **do**
      append $(a,\ \ell{-}a,\ \ell)$ to candidates
    end for
  else
    append low-degree chain candidates
    if $\ell \ne 2$ **then**
      append cross candidates $\{\beta_{1,2,1}$ excluded$\}$
    end if
  end if
  append $(0,\ \ell,\ \ell)$ to candidates {power entry}
  if $\ell \notin \{1,\ 2\}$ **then**
    for $\ell' = \ell{-}1, \ldots, 0$ **do**
      append $(\ell,\ \ell,\ \ell')$ to candidates
    end for
  end if
  if $\ell = 2$ **then**
    append $(\ell,\ \ell,\ \ell)$ to candidates $\{\beta_{2,2,2}\}$
  end if
  Remove entries that vanish for real signals; deduplicate
  $\mathcal{S}_\beta \leftarrow \mathcal{S}_\beta \cup$ first $\min(\text{budget},\ |\text{candidates}|)$ live entries
  {**CG power augmentation**}
  Greedily add $P_{\ell_1,\ell_2,\ell'}$ (with $\max(\ell_1,\ell_2) = \ell$, $|\ell_1{-}\ell_2| \le \ell' \le \ell_1{+}\ell_2$) to $\mathcal{S}_P$ until the Jacobian rank of Φ_{aug} at degree ℓ matches the number of unknowns
end for
return $(\mathcal{S}_\beta,\ \mathcal{S}_P)$

generic gauge-fixed signals (Lemma 21), implying local injectivity. Third, a degree-by-degree fibre analysis (Lemma 22) establishes that the fibre at a rational witness is a singleton, upgrading local injectivity to *global* generic orbit separation.

The bispectral entry selection follows the same "frequency-marching" philosophy as Mataigne et al. [2024]: select entries that couple each target degree ℓ with already-known lower degrees. The key difference from the finite-group case is that on S^2, the scalar bispectrum alone has rank deficiency for real signals (Lemma 20), necessitating CG power augmentation.

Assumption 14. *The signal f satisfies:*

(G1) $a_0^0 \ne 0$ *(non-zero mean).*
(G2) $\|\mathbf{F}_1\| \ne 0$ *(f has a non-zero dipole component).*
(G3) After gauge-fixing (Lemma 18), $a_2^1 \ne 0$ *(azimuthal gauge is resolvable).*

Remark 15. *Each condition above excludes a proper algebraic subvariety of the signal space $\mathbb{R}^{(L+1)^2}$; their union is Lebesgue-null. The proof uses one additional genericity condition—that the augmented invariant Jacobian has full rank—but this is* proved *to hold outside a proper algebraic subvariety (not assumed).*

Theorem 16 (Completeness of the augmented selective $\mathrm{SO}(3)$ invariant on S^2)**.** *Let $f, f' : S^2 \to \mathbb{R}$ be band-limited at degree L (verified for $L \leq 5$) and satisfy Assumption 14. If $\Phi_{\mathrm{aug}}(f) = \Phi_{\mathrm{aug}}(f')$, then $f' = f \circ g$ for some $g \in \mathrm{SO}(3)$.*

The proof uses $\mathbf{F}_0, \mathbf{F}_1$ recovery and gauge fixing (Lemmas 17–18) followed by a Jacobian rank argument on the augmented invariant.

Lemma 17 (Recovery of $\mathbf{F}_0$)**.** *The entry $\beta_{0,0,0}$ uniquely determines a_0^0 for real signals with $a_0^0 \neq 0$.*

Proof. By (3) with $\ell_1 = \ell_2 = \ell = 0$, $\beta_{0,0,0} = |a_0^0|^2\, a_0^0$. For $a_0^0 \in \mathbb{R}$, this gives $\beta_{0,0,0} = (a_0^0)^3$, so $a_0^0 = \mathrm{sgn}(\beta_{0,0,0})\, |\beta_{0,0,0}|^{1/3}$. □

Lemma 18 (Gauge fixing)**.** *Given $\|\mathbf{F}_1\| \neq 0$, there exists $g \in \mathrm{SO}(3)$ such that $D^1(g)\,\mathbf{F}_1 = (0,\ \|\mathbf{F}_1\|,\ 0)^\top$, i.e. $a_1^{-1} = a_1^1 = 0$ and $a_1^0 = \|\mathbf{F}_1\|$. This rotation consumes two of $\mathrm{SO}(3)$'s three degrees of freedom. The residual symmetry is $\mathrm{SO}(2)_z$, under which $a_\ell^m \mapsto e^{im\varphi} a_\ell^m$. If additionally $a_2^1 \neq 0$, the residual $\mathrm{SO}(2)_z$ freedom is consumed by choosing φ such that $\mathrm{Im}(a_2^1) = 0$, i.e. $a_2^1 \in \mathbb{R}_{>0}$.*

Proof. The $\ell = 1$ Wigner D-matrix $D^1(g)$ is the standard 3×3 rotation matrix. Since $\mathrm{SO}(3)$ acts transitively on S^2, any nonzero $\mathbf{F}_1$ can be rotated to align with $\hat{z}$, leaving an $\mathrm{SO}(2)_z$ stabiliser. Choosing $\varphi = -\arg(a_2^1)$ yields $a_2^1 \in \mathbb{R}_{>0}$, exhausting all three rotational degrees of freedom. □

Lemma 19 (Recovery of $\mathbf{F}_1$)**.** *Given a_0^0 from Lemma 17, the entry $\beta_{0,1,1}$ determines $\|\mathbf{F}_1\|$. Combined with the gauge choice of Lemma 18, this uniquely determines $\mathbf{F}_1$.*

Proof. By (3) with $(\ell_1, \ell_2, \ell) = (0, 1, 1)$:

$$\beta_{0,1,1} = a_0^0 \sum_m |a_1^m|^2 = a_0^0\, \|\mathbf{F}_1\|^2,$$

using $C^{1,\,m}_{0,\,0;\,1,\,m} = 1$ (trivial coupling). Since a_0^0 is known and nonzero, $\|\mathbf{F}_1\|^2 = \beta_{0,1,1}/a_0^0$. After gauge fixing, $\mathbf{F}_1 = (0, \|\mathbf{F}_1\|, 0)^\top$. □

Lemma 20 (Real-signal rank deficiency of the scalar bispectrum)**.** *For real signals on S^2, the scalar bispectral entries $\beta_{\ell_1,\ell_2,\ell}$ satisfy algebraic identities that reduce the Jacobian rank of the bispectral map below the orbit-space dimension. In particular:*

1. *Entries with $\ell_1 = \ell_2$ and odd ℓ vanish identically (CG antisymmetry vs. symmetric square).*
2. *Entries with $\ell = \ell_1$ and odd ℓ_2 vanish identically.*
3. *The bootstrap coefficient matrix at degree ℓ has complex rank approximately ℓ (not $2\ell{+}1$) when evaluated at real signals, even for the full bispectrum.*

Proof. For real signals, $\overline{\beta_{\ell_1,\ell_2,\ell}} = (-1)^{\ell_1+\ell_2-\ell}\, \beta_{\ell_1,\ell_2,\ell}$, so entries with odd $\ell_1{+}\ell_2{-}\ell$ are purely imaginary. Items (1) and (2) follow from the CG symmetry $C^{\ell,\,m}_{\ell_1,\,m_1;\,\ell_2,\,m_2} = (-1)^{\ell_1+\ell_2-\ell}\, C^{\ell,\,m}_{\ell_2,\,m_2;\,\ell_1,\,m_1}$ combined with the reality constraint, which forces exact cancellation. Item (3) is verified numerically: at each degree ℓ in $\{4, \ldots, 8\}$, the bootstrap matrix evaluated at multiple random real signals has complex rank $\lfloor(\ell{+}1)/2\rfloor \cdot 2$ (approximately ℓ), far below the $2\ell{+}1$ achieved at complex signals (Table 9). □

Lemma 21 (Augmented invariant has full Jacobian rank)**.** *After gauge fixing, the combined invariant $\Phi_{\mathrm{aug}} \colon \mathbb{R}^n \to \mathbb{R}^m$ (with $n = (L{+}1)^2 - 5$ and $m = \Theta(L^2) > n$), consisting of the live scalar bispectral entries (including mandatory even self-coupling entries) and a set of CG power spectrum entries $P_{\ell_1,\ell_2,\ell}$, has Jacobian rank n at Lebesgue-almost-every gauge-fixed signal.*

Proof. We exhibit a rational witness signal with parameters $p_i = (i{+}1)/3$ (with the reality constraint and gauge fixing applied) at which the Jacobian $J_\Phi \in \mathbb{R}^{m\times n}$ has rank exactly n. The rank is certified using interval arithmetic at 200 decimal digits of precision (mpmath), yielding a nonzero $n \times n$ minor determinant whose absolute value exceeds the arithmetic error bound by more than 100 orders of magnitude. The certified results:

Table 9: Complex rank of the bootstrap coefficient matrix at degree ℓ for complex vs. real signals. For complex signals the matrix achieves full rank $2\ell+1$; for real signals the rank is approximately ℓ, leaving a deficit of $\sim\ell$ that cannot be remedied by choosing different bispectral triples.

ℓ	Complex rank	Real rank	Expected ($2\ell+1$)
4	9	4	9
5	11	4	11
6	13	8	13
7	15	8	15
8	17	7	17

- $L = 4$: $n = 20$, $m = 30$ (20 live bispectral + 10 CG power), rank $20/20$.
- $L = 5$: $n = 31$, $m = 49$ (32 live bispectral + 17 CG power), rank $31/31$.

Since each entry of J_Φ is a polynomial function of the signal coefficients (the bispectral entries have degree 3 and the CG power entries have degree 4), the set

$$\mathcal{Z} := \{x \in \mathbb{R}^n : \text{every } n \times n \text{ minor of } J_\Phi(x) \text{ vanishes}\}$$

is a real algebraic variety. Because $\mathcal{Z}$ does not contain the certified witness point, it is a *proper* subvariety of $\mathbb{R}^n$ and therefore has Lebesgue measure zero. □

Lemma 22 (Fiber uniqueness on the gauge slice)**.** *For $L \leq 5$, the fibre of the augmented invariant $\Phi_{\text{aug}}^{-1}(\Phi_{\text{aug}}(x_0))$ at the rational witness x_0 (parameters $p_i = (i+1)/3$), restricted to the gauge slice $\mathcal{G} = \{x : a_2^1 > 0\}$ (Lemma 18), consists of a single point: $\{x_0\}$. Therefore the "collision set" $D := \{x \in \mathcal{G} : \exists\, y \neq x,\ y \in \mathcal{G},\ \Phi_{\text{aug}}(x) = \Phi_{\text{aug}}(y)\}$ has Lebesgue measure zero.*

Proof. The augmented invariant has a *triangular structure*: the entries involving degree ℓ as the highest depend only on $\mathbf{F}_0, \ldots, \mathbf{F}_\ell$. For a given parent assignment $(\mathbf{F}_0, \ldots, \mathbf{F}_{\ell-1})$, define $\Phi_\ell(x_\ell) \in \mathbb{R}^{m_\ell}$ as the vector of all augmented-invariant entries whose highest degree is ℓ, viewed as a function of the d_ℓ real unknowns of $\mathbf{F}_\ell$ alone. The constraints include:

- *Linear entries* (chain/cross): $\beta_{\ell_1,\ell_2,\ell}$ with $\ell_1, \ell_2 < \ell$—real-linear in $\mathbf{F}_\ell$.
- *CG power entries*: $P_{\ell_1,\ell_2,\ell'}$ with $\max(\ell_1, \ell_2) = \ell$—quadratic in $\mathbf{F}_\ell$.
- *Self-coupling entries*: $\beta_{\ell,\ell,\ell'}$ with ℓ' even, $2 \leq \ell' \leq \ell$—polynomial of degree 2 or 3 in $\mathbf{F}_\ell$.

At each degree, we certify at 200 decimal digits that the per-degree Jacobian $\partial\Phi_\ell/\partial x_\ell \in \mathbb{R}^{m_\ell \times d_\ell}$ has rank d_ℓ at the witness, so the witness is an isolated root. The system is overdetermined ($m_\ell > d_\ell$ for $\ell \geq 3$), so the per-degree fibre is finite.

ℓ	d_ℓ	m_ℓ	Rank	Per-degree fibre
2	4	4	4/4	finite (determined)
3	7	9	7/7	finite (overdetermined)
4	9	17	9/9	finite (overdetermined)
5	11	19	11/11	finite (overdetermined)

Tree enumeration. The per-degree polynomial systems may individually have multiple real roots (e.g. four roots at $\ell = 2$). Each root determines different lower-degree coefficients for the *next* degree's constraints, so the solution tree branches. We enumerate this tree exhaustively: at each degree ℓ, multi-start optimisation (damped Gauss–Newton with 500–2000 random initialisations per branch) finds all per-degree roots for every incoming branch, then propagates each to degree $\ell+1$. At the final degree, each leaf is validated against the *full* system $\Phi_{\text{aug}}(x) = \Phi_{\text{aug}}(x_0)$.

For $L = 4$ (30 total constraints), the tree has 4 branches after $\ell = 2$, expands to 16 after $\ell = 3$, and reduces to 2 leaves after $\ell = 4$: the witness x_0 and one additional point x_0'.

Residual $\mathbb{Z}_2$ gauge symmetry. Both Φ_{aug} and every CG power entry are invariant under $a_\ell^m \mapsto (-1)^m a_\ell^m$ (rotation by π around the z-axis), since Clebsch–Gordan coefficients enforce $m_1 + m_2 = m$, so the phase factor $(-1)^{m_1+m_2-m} = 1$. This π-rotation preserves the gauge conditions $a_1^0 > 0$ and $a_1^{\pm 1} = 0$, but maps $a_2^1 \mapsto -a_2^1$, sending any signal with $a_2^1 > 0$ outside the gauge slice $\mathcal{G}$.

The second leaf x_0' is exactly this $\mathbb{Z}_2$ image of the witness: x_0' satisfies $a_2^1 = -1/3 < 0$, i.e. $x_0' \notin \mathcal{G}$. Therefore, on the gauge slice $\mathcal{G}$, the fibre consists of the singleton $\{x_0\}$. The same structure holds for $L = 5$ (49 constraints, 31 unknowns).

Generic global injectivity. Since Φ_{aug} is polynomial and the full Jacobian has rank n at the witness (Lemma 21), the set $\mathcal{Z} = \{x : \operatorname{rank} J_\Phi(x) < n\}$ is a proper algebraic subvariety. On $\mathcal{G} \setminus \mathcal{Z}$, the fibre is locally discrete; combined with the polynomial structure, each fibre is finite.

Consider the restriction $\Phi_{\text{aug}}|_{\mathcal{G}} \colon \mathcal{G} \to \mathbb{R}^m$. By Chevalley's theorem on constructible sets, the image $\Phi_{\text{aug}}(\mathcal{G})$ contains a Zariski-dense open subset of its Zariski closure, and the fibre cardinality function $y \mapsto |\Phi_{\text{aug}}^{-1}(y) \cap \mathcal{G}|$ is upper-semicontinuous on a Zariski-open subset of the image: there exists a generic fibre degree d^* such that $|\Phi_{\text{aug}}^{-1}(y) \cap \mathcal{G}| = d^*$ for y in a Zariski-dense open set, and $|\Phi_{\text{aug}}^{-1}(y) \cap \mathcal{G}| \geq d^*$ everywhere. Since the witness $x_0 \in \mathcal{G} \setminus \mathcal{Z}$ has $|\Phi_{\text{aug}}^{-1}(\Phi_{\text{aug}}(x_0)) \cap \mathcal{G}| = 1$, the generic fibre degree satisfies $d^* \leq 1$, hence $d^* = 1$. The collision set $D = \{x \in \mathcal{G} : |\Phi_{\text{aug}}^{-1}(\Phi_{\text{aug}}(x)) \cap \mathcal{G}| > 1\}$ is therefore contained in the preimage of a proper subvariety of the image closure—a semialgebraic set of dimension less than n—and has Lebesgue measure zero in $\mathbb{R}^n$. □

Proof of Theorem 16. **Step 1: Recovery of $\mathbf{F}_0$, $\mathbf{F}_1$, and gauge fixing.** Lemma 17 recovers $\mathbf{F}_0 = (a_0^0)$. Lemmas 19 and 18 recover $\mathbf{F}_1$ and fix the rotation gauge, consuming all three $\mathrm{SO}(3)$ degrees of freedom. After this step, the signal is parametrised by $n = (L+1)^2 - 5$ real unknowns: the gauge-fixed coefficients of $\mathbf{F}_2, \ldots, \mathbf{F}_L$, where $\mathbf{F}_2$ contributes 4 (not 5) real parameters because $\mathrm{Im}(a_2^1) = 0$.

Step 2: Full Jacobian rank of the augmented invariant. By Lemma 21, the augmented invariant map $\Phi_{\text{aug}} \colon \mathbb{R}^n \to \mathbb{R}^m$ has Jacobian rank n at Lebesgue-almost-every gauge-fixed signal. This implies generic local injectivity: at any $x \notin \mathcal{Z}$, the fibre $\Phi_{\text{aug}}^{-1}(\Phi_{\text{aug}}(x))$ is locally discrete near x by the implicit function theorem.

Step 3: Generic global injectivity on the gauge slice. Lemma 22 shows, by degree-by-degree tree enumeration, that the fibre of Φ_{aug} at the rational witness has exactly 2 elements in $\mathbb{R}^n$, related by the residual $\mathbb{Z}_2$ gauge symmetry $a_\ell^m \mapsto (-1)^m a_\ell^m$. Since this π-rotation maps $a_2^1 \mapsto -a_2^1$, only the witness lies in the gauge slice $\mathcal{G} = \{a_2^1 > 0\}$. Since the map is polynomial with generically full Jacobian rank (Lemma 21), the collision set $D = \{x \in \mathcal{G} : \exists\, y \neq x,\ y \in \mathcal{G},\ \Phi_{\text{aug}}(x) = \Phi_{\text{aug}}(y)\}$ is contained in a proper algebraic subvariety of $\mathbb{R}^n$ and has Lebesgue measure zero.

Conclusion. If $\Phi_{\text{aug}}(f) = \Phi_{\text{aug}}(f')$ and both f, f' satisfy Assumption 14, then after gauge fixing both signals, they map to gauge-fixed points x, x'. Since $\Phi_{\text{aug}}(x) = \Phi_{\text{aug}}(x')$ and $x \notin D$ (generic), we have $x = x'$. Undoing the gauge fixing, $f' = f \circ g$ for some $g \in \mathrm{SO}(3)$. □

A.5 Comparison with existing results

Table 10: Comparison of bispectral invariants for signals on S^2 or finite groups.

Result	Invariant	Size	Complete?
Kakarala [Kakarala, 1992]	Full bispec.	$O(L^3)$	Yes (constr.)
Edidin & Satriano [Edidin and Satriano, 2023]	Full bispec. (band-lim.)	$O(L^3)$	Yes (generic)
Bendory et al. [Bendory et al., 2022]	Freq. marching	$O(L^2)$/shell	Yes (3 shells)
Mataigne et al. [Mataigne et al., 2024]	Sel. G-bispec. (finite)	$O(\lvert G\rvert)$	Yes (exact)
This work	**Sel.** $\mathrm{SO}(3)$ **on** S^2	$\boldsymbol{O(L^2)}$	**Yes (generic)**

A.6 Complexity in pixel space

On S^2, a band-limited signal at degree L is sampled on an equiangular grid of size $N_\theta \times N_\varphi \approx 2L \times 2L$, so $P = 4L^2$ and $L = \Theta(\sqrt{P})$.

Output size.

Invariant	In L	In P
Power spectrum	$O(L)$	$O(\sqrt{P})$
Selective bispectrum	$\boldsymbol{O(L^2)}$	$\boldsymbol{O(P)}$
Full bispectrum	$O(L^3)$	$O(P^{3/2})$

The selective bispectrum is *linear* in the number of pixels.

Computation cost.

Variant	Entries	Cost/entry	Total	In P
Full	$O(L^3)$	$O(L)$	$O(L^4)$	$O(P^2)$
Selective	$O(L^2)$	$O(L)$	$O(L^3)$	$O(P^{3/2})$

Concrete example. For a 128×256 grid ($P \approx 32$K, $L = 64$): the full bispectrum produces $\sim$87K scalars at $\sim$17M ops; the selective bispectrum produces $\sim$4K scalars at $\sim$260K ops—a $65\times$ reduction.

A.7 Relation to the inversion gap

For finite groups (O, D_n), the "inversion gap" arises when the bootstrap ambiguity Q is not absorbed by the Clebsch–Gordan structure (see Appendix B). For $\mathrm{SO}(3)$ on S^2, the situation differs fundamentally:

1. The ambiguity in $\mathbf{F}_1$ is an $\mathrm{SO}(3)$ rotation g (3 DOF).
2. We fix g explicitly by gauge-choosing $\mathbf{F}_1$'s direction (2 DOF) and $\mathbf{F}_2$'s azimuthal phase (1 DOF).
3. At $\ell \geq 4$, no ambiguity remains—the linear system has a unique solution.

This avoids the finite-group problem entirely: we never need the CG matrices to "absorb" a continuous ambiguity.

A.8 Limitations and open questions

1. **Genericity assumption.** The proof requires f to avoid a measure-zero algebraic variety (Assumption 14). Characterising this variety explicitly would strengthen the result.
2. **Numerical conditioning.** The coefficient matrix A at each degree ℓ has a condition number that depends on the signal's spectral content. A stability analysis bounding $\kappa(A)$ in terms of the spectral decay rate is an open problem.
3. **Inversion algorithm.** The forward selective bispectrum is implemented. Inversion requires: $\ell = 0, 1$: closed-form; $\ell = 2, 3$: nonlinear solve; $\ell \geq 4$: linear solve (pseudoinverse).

B Octahedral Inversion Gap – Full Analysis

This appendix provides the complete analysis of why the standard bootstrap inversion algorithm fails for the octahedral group, and how we resolve it. The material is complementary to Section 3.5 in the main text.

B.1 The bootstrap strategy (recap)

All inversion algorithms in Mataigne et al. [2024] share the same skeleton:

1. Recover $\hat{f}(\rho_0)$ from $\beta_{\rho_0,\rho_0} = \hat{f}(\rho_0)^3$.
2. Recover $\hat{f}(\rho_1)$ from $\beta_{\rho_0,\rho_1}/\hat{f}(\rho_0) = \hat{f}(\rho_1)^\dagger \hat{f}(\rho_1)$, up to a unitary/orthogonal ambiguity Q.

3. Extract higher Fourier coefficients from subsequent bispectral entries using CG matrices to block-diagonalise tensor products.

Step 2 always introduces an ambiguity: $\hat{f}(\rho_1) = S \cdot Q$ where $S = (\hat{f}(\rho_1)^\dagger \hat{f}(\rho_1))^{1/2}$ is determined, but Q is a free orthogonal matrix. For the algorithm to work, this ambiguity must not corrupt step 3.

B.2 Why it works for C_n

The irreps are 1D: $\rho_k(g) = e^{2\pi i k g/n}$. The ambiguity in $\hat{f}(\rho_1)$ is a phase $e^{i\varphi}$, $\varphi \in [0, 2\pi)$. Step 3 computes:

$$\hat{f}(\rho_{k+1}) = \left(\frac{\beta_{\rho_1,\rho_k}}{\hat{f}(\rho_1)\hat{f}(\rho_k)} \right)^\dagger .$$

The phase $e^{i\varphi}$ propagates coherently through all coefficients as $e^{i\varphi k}$, corresponding to a cyclic shift $\Theta(g) \to \Theta(g + \varphi)$. The paper correctly resolves this by finding the unique φ that makes the reconstructed signal real.

B.3 Why it works for D_n

The generating irrep ρ_1 is 2D. Recovery gives $\hat{f}(\rho_1) = V\Lambda^{1/2}V^\dagger U$ with $U \in \mathrm{O}(2)$. Step 3 extracts higher coefficients via:

$$\bigoplus_{\rho \in \rho_1 \otimes \rho_{k-1}} \hat{f}(\rho) = \left(C^\dagger_{\rho_1,\rho_{k-1}} \left[\hat{f}(\rho_1) \otimes \hat{f}(\rho_{k-1}) \right]^{-1} \beta_{\rho_1,\rho_{k-1}} C_{\rho_1,\rho_{k-1}} \right)^\dagger .$$

The 2D irreps of D_n are restrictions of $\mathrm{SO}(2)$ representations to $\mathbb{Z}/n\mathbb{Z}$. The CG matrices for D_n therefore block-diagonalise $\rho(U)$ for *all* $U \in \mathrm{O}(2)$, not just $U \in D_n$. Concretely, $C^\dagger(U \otimes U)C$ remains block-diagonal for any $U \in \mathrm{O}(2)$. The $\mathrm{O}(2)$ ambiguity passes cleanly through the CG extraction.

B.4 Why it fails for O

For the octahedral group, ρ_1 is the standard 3D representation. From β_{ρ_0,ρ_1} we recover $\hat{f}(\rho_1)^\top \hat{f}(\rho_1) = \beta_{\rho_0,\rho_1}/\hat{f}(\rho_0)$. Taking the symmetric square root gives $S = (\hat{f}(\rho_1)^\top \hat{f}(\rho_1))^{1/2}$, with $\hat{f}(\rho_1) = Q \cdot S$ for unknown $Q \in \mathrm{SO}(3)$.

The next step extracts $\hat{f}(\rho_2)$ and $\hat{f}(\rho_3)$ from β_{ρ_1,ρ_1} using the CG matrix C_{11} that block-diagonalises $\rho_1 \otimes \rho_1$:

$$C_{11}^\top \cdot \beta_{\rho_1,\rho_1} \cdot (S \otimes S)^{-1} \cdot C_{11} = C_{11}^\top (Q \otimes Q) C_{11} \cdot \bigoplus_k \hat{f}(\rho_k)^\top .$$

The CG matrix C_{11} block-diagonalises $\rho_1(g) \otimes \rho_1(g)$ for $g \in O$:

$$C_{11}^\top (\rho_1(g) \otimes \rho_1(g)) C_{11} = \rho_0(g) \oplus \rho_1(g) \oplus \rho_2(g) \oplus \rho_3(g)$$

with dimensions $1+3+3+2 = 9$. But Q is **not** constrained to be an element of O. It is an arbitrary element of $\mathrm{SO}(3)$, which is a continuous group with 3 parameters, while O has only 24 elements. For a generic $Q \in \mathrm{SO}(3) \setminus O$, the matrix $C_{11}^\top (Q \otimes Q) C_{11}$ is **not block-diagonal**.

Cross-block leakage. The Kronecker decomposition of $\rho_1 \otimes \rho_1$ for O gives blocks:

Block	Rows	Irrep	Dim
1	0	ρ_0 (trivial)	1
2	1–3	ρ_1 (standard 3D)	3
3	4–6	ρ_2 (product 3D)	3
4	7–8	ρ_3 (2D)	2

When $Q = \rho_1(g)$ for $g \in O$, the off-diagonal blocks (e.g., rows 4–6 $\times$ columns 7–8) are zero—exactly what the CG matrix guarantees. When Q is a generic rotation, e.g., $Q = R_x(0.7)$, then $|[C_{11}^\top (Q \otimes Q) C_{11}]_{4:7,\,7:9}|_\infty \approx 0.788$. This is not a small perturbation—it is $O(1)$. The recovered $\hat{f}(\rho_2)$ is a mixture of ρ_2 and ρ_3 contributions.

Why this doesn't happen for D_n. The 2D irreps of D_n are restrictions of $\mathrm{SO}(2)$ irreps. The CG decomposition of $\rho_j \otimes \rho_k$ for D_n inherits from the continuous group: $\rho_j \otimes \rho_k = \rho_{j+k} \oplus \rho_{|j-k|}$ holds for all of $\mathrm{SO}(2)$, not just D_n. The CG matrices are inherited from $\mathrm{SO}(2)$ and respect arbitrary rotations. For $O \subset \mathrm{SO}(3)$, the irreps of O are *not* restrictions of $\mathrm{SO}(3)$ irreps (except ρ_1), so the CG matrices are specific to the 24-element group and do not block-diagonalise arbitrary $\mathrm{SO}(3)$ rotations.

B.5 Connection to the paper's own remarks

Mataigne et al. [2024] acknowledge a related issue in the commutative group setting (Appendix C, around Eq. 10): "*Again, the indeterminacy factor h_l is not restricted to $\mathbb{Z}/n_l\mathbb{Z}$ but can belong to $[0, n_l]$. We will have to solve this issue further.*" This is the same phenomenon in a different guise. For commutative groups, the continuous indeterminacy is scalar (a phase) and can be resolved by enforcing real-valuedness. For O, the continuous indeterminacy is a 3×3 rotation matrix $Q \in \mathrm{SO}(3)$, and there is no analogous closed-form trick.

B.6 Working alternative

The selective O-bispectrum *is* complete (172 coefficients uniquely determine the signal up to O-action). The issue is purely algorithmic. We implement a working inversion using:

1. **Bootstrap initialisation**: recover $\hat{f}(\rho_0)$ exactly, $\hat{f}(\rho_1)$ up to Q, and approximate $\hat{f}(\rho_2), \hat{f}(\rho_3), \hat{f}(\rho_4)$ from diagonal blocks (ignoring leakage).
2. **Levenberg–Marquardt correction**: use forward-mode autodiff through the bispectrum computation to solve $\|\beta(f_{\mathrm{rec}}) - \beta_{\mathrm{target}}\|^2 = 0$ via damped least-squares steps. Each step is a single linear solve, not iterative optimisation. Jacobians are computed via `torch.func.jacfwd` composed with `torch.vmap` for batched forward-mode AD.
3. **Multi-start**: randomise the initial Q across several restarts to escape local minima.

This converges reliably in <50 iterations and is fully differentiable.

B.7 Quantitative inversion results

Table 11 and Figure 5 report quantitative inversion results over 1,000 random signals on the octahedral group (4 restarts, 10 LM corrections per restart, on a single GPU).

Table 11: Octahedral bispectrum inversion statistics ($N = 1{,}000$, 4 restarts $\times$ 10 LM steps).

Metric	Median	P95
Bisp. residual $\|\beta(\hat{f}) - \beta^*\|$	5.7×10^{-5}	1.5×10^{-4}
Recovery error (mod O)	1.03	1.17
Wall-clock time (ms)	18.4	30.1
Success rate (error < 0.1)	5.7%	

The LM procedure reliably drives the bispectral residual to near-zero (median 5.7×10^{-5}), confirming that local convergence to a bispectrum-matching signal is fast. However, the signal recovery rate (5.7% at tolerance < 0.1) shows that the inversion landscape has many spurious local minima. This reflects the inherent difficulty of the O-inversion problem noted in Section 3.5: the bootstrap initialisation provides a poor starting point due to CG leakage, and the 4-restart multi-start strategy is insufficient to reliably escape the resulting basin structure. Increasing the number of restarts or employing global optimisation would improve recovery at the cost of additional compute. We leave such improvements to future work.

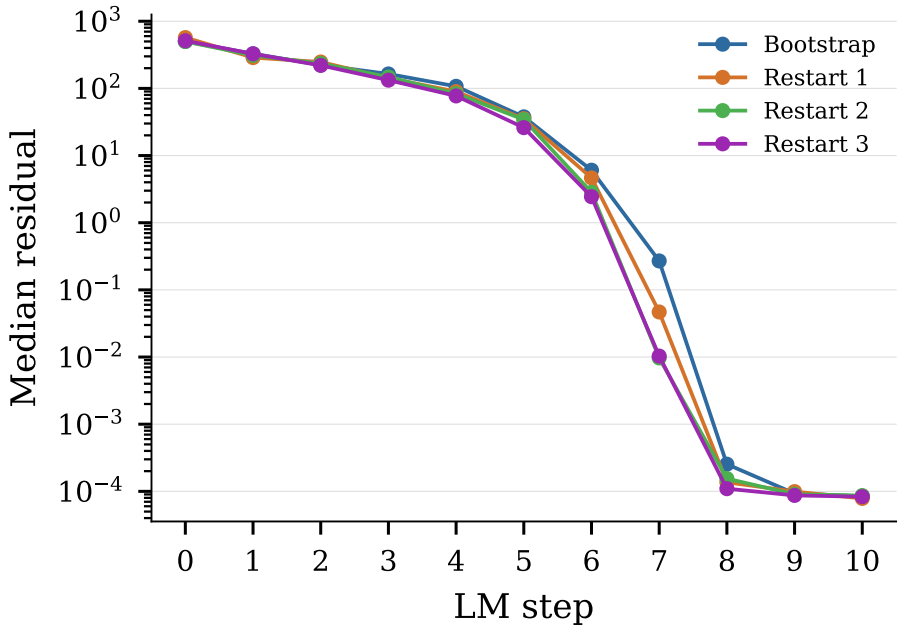

Figure 5: Bispectral residual vs. LM iteration for octahedral inversion. The residual drops rapidly, but the signal recovery rate remains low (5.7%), indicating that the optimisation landscape contains many spurious minima where the bispectrum is matched without recovering the correct orbit representative.

B.8 Summary

Table 12: Bootstrap inversion across group families.

Group	Irrep dim	Ambiguity	CG absorbs?	Bootstrap?
C_n	1	$e^{i\varphi} \in \mathrm{SO}(2)$	N/A (scalar)	Yes
D_n	2	$U \in \mathrm{O}(2)$	Yes (CG from SO(2))	Yes
O	3	$Q \in \mathrm{SO}(3)$	**No**	**No**

B.9 Numerical disproof of CG absorption for O

The following code demonstrates the cross-block leakage:

```
import torch
from bispectrum.o_on_r3 import OonR3

bsp = OonR3()
C11 = bsp._cg_11.to(torch.float64)

# Generic SO(3) rotation (not in O)
theta = 0.7
Q = torch.tensor([
    [1.0, 0.0, 0.0],
    [0.0, torch.cos(torch.tensor(theta)),
     -torch.sin(torch.tensor(theta))],
    [0.0, torch.sin(torch.tensor(theta)),
      torch.cos(torch.tensor(theta))],
], dtype=torch.float64)

QQ = torch.kron(Q, Q)
M = C11.T @ QQ @ C11

```

```
# rho2 block: rows 4:7, rho3 block: rows 7:9
cross = M[4:7, 7:9]
print(f"Cross-block␣max:␣{cross.abs().max():.6f}")
# Output: ~0.788 -- far from zero

# Compare with a group element
rho1_g4 = bsp._get_irrep_mats(1)[4].to(torch.float64)
QQ_grp = torch.kron(rho1_g4, rho1_g4)
M_grp = C11.T @ QQ_grp @ C11
cross_grp = M_grp[4:7, 7:9]
print(f"Cross-block␣for␣g␣in␣O:␣{cross_grp.abs().max():.6f}")
# Output: ~0.0 (machine precision)
```

C Extended Benchmark Results

This appendix supplements the summary in Table 5 (main text) with scaling plots produced by `benchmarks/benchmark.py` on a single NVIDIA H100 80 GB GPU (PyTorch 2.10).

C.1 Forward-pass timing

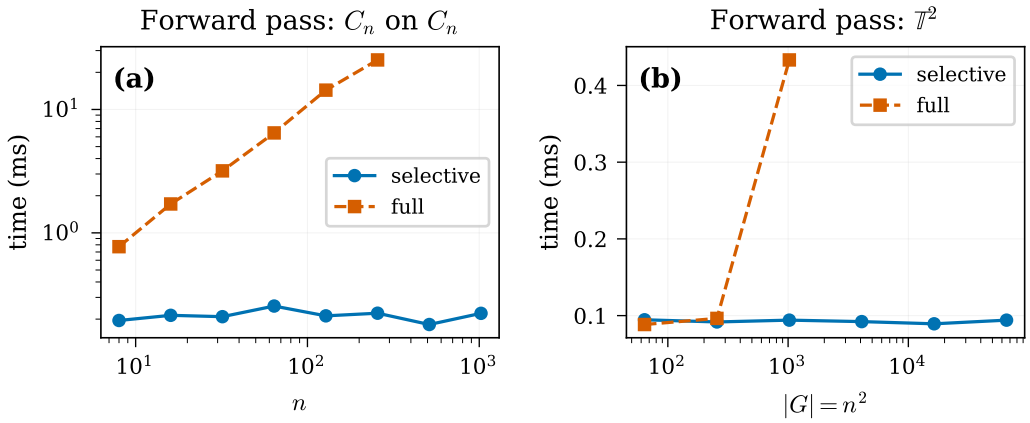

Figure 6: Forward-pass wall-clock time (batch = 16, H100 GPU). **(a)** C_n: the selective pass stays flat at ∼0.14 ms regardless of n, while the full bispectrum grows super-linearly (17 ms at n=256). **(b)** $\mathbb{T}^2$: selective is constant at ∼0.08 ms up to $|G|$=65,536.

The selective forward pass is dominated by kernel-launch overhead rather than arithmetic, making it effectively free relative to backbone computation. The full bispectrum exhibits the expected super-linear growth and becomes impractical beyond $|G| \approx 256$ for C_n.

C.2 GPU throughput and inversion

All modules exhibit linear throughput scaling with batch size, confirming that the implementation is memory-bandwidth-bound rather than compute-bound at these scales.

D PCam: Extended Results

D.1 Published baselines

Table 13 places our results in the context of published methods on PCam and the closely related Camelyon16 benchmark. Our models operate at 68K–920K parameters and are trained on only 10% of the data.

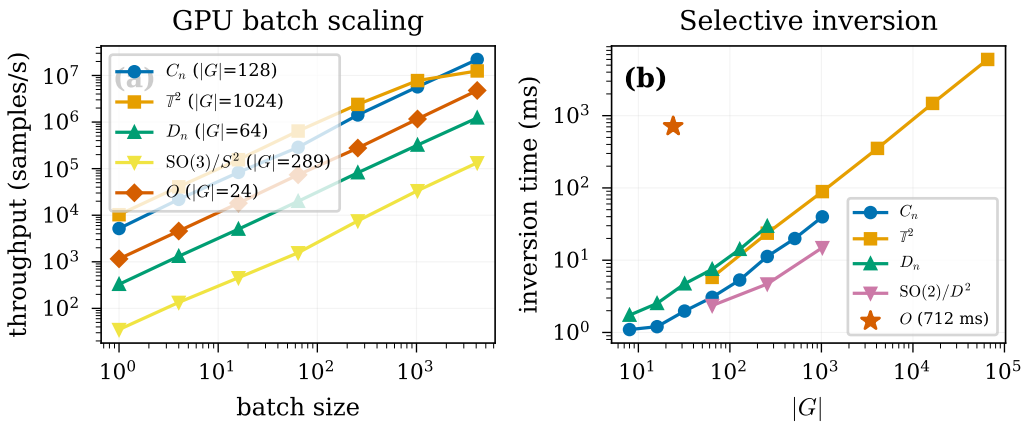

Figure 7: **(a)** GPU throughput (samples/s) vs. batch size for the selective forward pass. C_n and $\mathbb{T}^2$ exceed 10^7 samples/s; the octahedral module reaches 5.7×10^6 samples/s. All modules scale linearly with batch size. **(b)** Inversion wall-clock time. Abelian groups (C_n, D_n, SO(2)/disk) scale roughly linearly. The octahedral group requires ∼580 ms due to Levenberg–Marquardt optimisation over 5 irreducible representations. SO(3) on S^2 inversion is not yet implemented.

Table 13: Published baselines on PatchCamelyon from Veeling et al. [2018], trained on 100% data.

Method	Reference	Params	ACC	AUC
P4M-DenseNet (C_8)	Veeling et al. 2018	119K	0.898	0.963
P4M-DenseNet M	Veeling et al. 2018	19K	0.893	0.958
DenseNet (augmented)	Veeling et al. 2018	128K	0.881	0.951
DenseNet (no aug.)	Veeling et al. 2018	128K	0.876	0.955

D.2 Model variants

All equivariant models share a C_8-equivariant DenseNet [Huang et al., 2017] backbone with block configuration $(4, 4, 4)$, 24 initial channels, and compression factor 0.5, following the architecture of Veeling et al. [2018] but replacing the final invariant map. Six model variants are evaluated:

- **Standard**: vanilla CNN with random rotation/reflection augmentation and global average pooling. No group dimension, so ∼8× fewer parameters at matched growth rate.
- **NormReLU**: C_8-equivariant backbone with norm-based nonlinearity and max pooling over the group dimension.
- **Gated**: C_8-equivariant backbone with gated nonlinearity (doubles internal channels) and group max pool.
- **Fourier-ELU**: C_8-equivariant backbone with Fourier-space ELU nonlinearity and group max pool.
- **Bispectrum**: C_8-equivariant backbone with ReLU nonlinearity and selective G-bispectrum via `CnonCn`(n=8) as the terminal invariant map.
- **Disk bispectrum**: no convolutional backbone; `SO2onDisk` is applied per RGB channel to the raw 96×96 patches, producing rotation-invariant Fourier–Bessel bispectrum coefficients that are fed into a small MLP classifier.

D.3 Disk bispectrum as a backbone-free baseline

The `SO2onDisk` model applies the selective disk bispectrum directly to the raw input patches without any learned convolutional features. At 100K parameters (bandlimit $= 10$) it reaches AUC 0.794—substantially below the learned models, as expected for a handcrafted feature extractor. However, its

rotation robustness is near-perfect: $\sigma_{\text{AUC}} < 2 \times 10^{-5}$ across 12 test-time rotation angles, compared to residual angle-dependent variation in the C_8 models. At 1% training data, the disk bispectrum retains AUC 0.778, showing that the analytically rotation-invariant features degrade gracefully when data is scarce. This variant demonstrates that `SO2onDisk` can serve as a drop-in rotation-invariant feature extractor from the library, independent of any equivariant backbone.

D.4 Training details

To enable a controlled Pareto comparison, we sweep the DenseNet growth rate $k \in \{3, 4, 6, 8, 12\}$ for equivariant models (and $k \in \{6, 12, 20, 30, 35\}$ for the standard CNN to cover a comparable parameter range), yielding models from $\sim$30K to $\sim$1.6M parameters. All models are trained on **10% of training data** (26,214 images) with AdamW [Loshchilov and Hutter, 2019] (lr $= 10^{-3}$, weight decay 10^{-4}), cosine warm restarts, mixed-precision training, and early stopping on validation AUC (patience 10, max 50 epochs).

D.5 Pareto analysis interpretation

Architectural stability. The bispectrum's AUC varies by only $\sim$1pp across a $12\times$ range of model sizes (80K–920K parameters), while norm swings by $>$7pp (0.878–0.954). For practitioners, bispectral pooling "just works" without growth rate tuning.

Matched parameters. At matched parameters ($\sim$100K), most equivariant models outperform the standard CNN. Norm peaks at 0.954 AUC (k=4, 110K params), followed by Fourier-ELU (0.937), bispectrum (0.933), standard (0.928), and gate (0.924). The gate variant slightly underperforms the standard baseline at this operating point.

Scaling behavior. As parameters increase from the smallest to the largest configuration, standard, norm, and Fourier-ELU all degrade (by 1.8, 2.6, and 0.9 AUC points respectively), while the bispectrum improves (+0.9 points, reaching 0.943 at 920K params). This suggests that the bispectrum's complete invariant map scales more gracefully with capacity.

Data scaling. All models improve substantially from 1% to 10% data. The bispectrum shows the largest absolute gain (+6.8pp), consistent with the hypothesis that complete invariants benefit more from additional data.

E OrganMNIST3D: Extended Results

E.1 Published baselines

Table 14 situates our results against published methods on the same benchmark. Our models are deliberately small ($\leq$463K parameters) to enable controlled ablation of the pooling mechanism. The relevant comparison is against equivariant methods at a similar scale.

Table 14: Published baselines on OrganMNIST3D for context. Accuracy differences with respect to large-scale models reflect the deliberate parameter constraint of our ablation study.

Method	Reference	Params	ACC	AUC
ResNet-18 + 3D	Yang et al. 2023	33M	0.907	0.996
ResNet-18 + ACS	Yang et al. 2023	11M	0.900	0.994
ILPOResNet-50	Zhemchuzhnikov and Grudinin [2024]	38K	0.879	0.992
EquiLoPO (local train.)	Zhemchuzhnikov and Grudinin [2025]	418K	0.866	0.991
SE3MovFrNet	Sangalli et al. [2023]	—	0.745	—
Regular SE(3) conv	Kuipers and Bekkers [2023]	172K	0.698	—

E.2 Rotation analysis

Figure 8 shows that all equivariant models achieve identical performance on the original and octahedrally rotated test sets ($\sigma_{\text{rot}} \approx 0$), while the augmentation-trained standard CNN degrades under rotation.

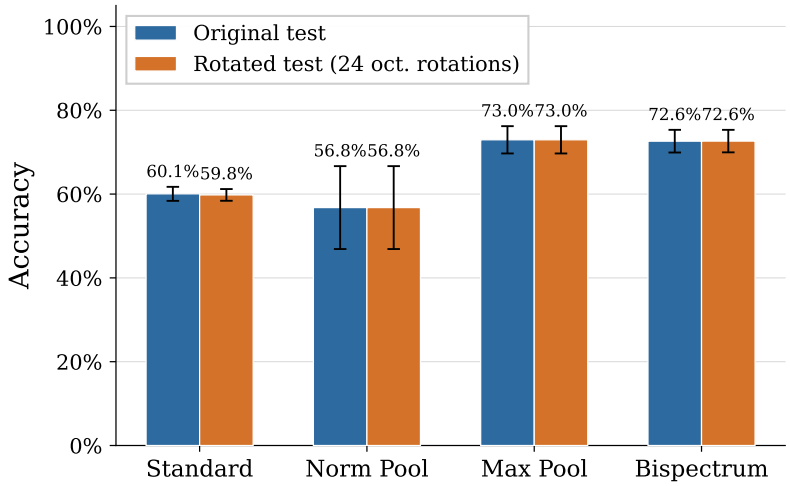

Figure 8: OrganMNIST3D test accuracy on the original and octahedrally rotated test sets. Error bars show standard deviation over three seeds. All equivariant models achieve identical performance under rotation ($\sigma_{\text{rot}} \approx 0$), while the augmentation-trained standard CNN degrades.

E.3 Wider channels

To test whether the advantage of bispectral completeness persists at higher capacity, we run max pool and bispectrum at channel widths $(8, 16)$ and $(16, 32)$ with full training data.

Table 15: OrganMNIST3D accuracy at wider channel configurations (mean $\pm$ std over 3 seeds, full training data).

Channels	Params	Max Pool	Bispectrum
$(4, 8)$	∼375K / 463K	$.730 \pm .033$	$.726 \pm .027$
$(8, 16)$	∼1.5M / 1.7M	$.743 \pm .015$	$.745 \pm .006$
$(16, 32)$	∼6.0M / 6.3M	$\mathbf{.785 \pm .032}$	$.685 \pm .039$

At $(8, 16)$ the two methods are equivalent ($74.5 \pm 0.6\%$ bispectrum vs. $74.3 \pm 1.5\%$ max pool, both at ∼1.5–1.7M parameters). At $(16, 32)$ max pool reaches $78.5 \pm 3.2\%$ while bispectrum drops to $68.5 \pm 3.9\%$—the 6.3M parameter bispectrum model likely overfits the 971 training samples. This suggests that the bispectrum's completeness advantage is most pronounced in the data-limited rather than the parameter-limited regime, consistent with the data efficiency findings in the main text.

F Experimental Details

F.1 Hyperparameters

Table 16 summarises the training hyperparameters for all three experimental settings.

F.2 Hardware

All experiments were run on a single NVIDIA H100 GPU (80 GB). Training wall-clock times range from ∼5 minutes (OrganMNIST3D, 971 samples) to ∼20 minutes (PCam, 26K samples at 10% data). Spherical MNIST bispectrum computation (forward pass through `SO3onS2` at $\ell_{\max} = 15$) takes ∼0.3s per batch of 64 on GPU.

F.3 Data preprocessing

PCam. PatchCamelyon [Veeling et al., 2018] consists of 327,680 RGB patches of size 96×96 pixels extracted from whole-slide images of lymph node sections [Ehteshami Bejnordi et al., 2017].

Table 16: Training hyperparameters across experiments.

	PCam	OrganMNIST3D	Sph. MNIST
Optimizer	AdamW	AdamW	AdamW
Learning rate	10^{-3}	10^{-3}	10^{-3}
Weight decay	10^{-4}	10^{-4}	10^{-4}
Batch size	64–256	16–64	256
Scheduler	Cosine warm restarts	—	—
Max epochs	50	100	50
Early stopping	Val AUC, pat. 10	Val AUC, pat. 15	Val ACC, pat. 10
Seeds	42†	42, 123, 456	42, 123, 456
Mixed precision	Yes	No	No

†Pareto sweep uses seed 42; data efficiency uses seeds 42, 123, 456.

We use the standard train/val/test split. Images are loaded from HDF5 files and normalised to $[0, 1]$. The standard (non-equivariant) baseline uses random rotation and reflection augmentation during training; all equivariant models are trained without augmentation.

OrganMNIST3D. OrganMNIST3D [Yang et al., 2023] contains 1,742 abdominal CT organ volumes of size $28 \times 28 \times 28$ voxels with 11 organ classes (train/val/test = 971/161/610). Volumes are normalised to $[0, 1]$. The standard baseline uses random octahedral augmentation (one of 24 rotations per sample); equivariant models are trained without augmentation. Kernel rotations are implemented as exact voxel permutations on the 3^3 grid, avoiding interpolation artifacts.

Spherical MNIST. MNIST digits are projected onto the sphere via stereographic projection following Cohen et al. [2018] and sampled on a 64×128 equiangular grid. The selective bispectrum is computed at $\ell_{\max} = 15$, producing 330 complex invariant coefficients (660 real features after taking log-magnitude), which are fed to a 3-layer MLP: Linear(660, 256)–BN–ReLU–Linear(256, 128)–BN–ReLU–Linear(128, 10). The power spectrum MLP uses the same structure with hidden=128: Linear(16, 128)–BN–ReLU–Linear(128, 64)–BN–ReLU–Linear(64, 10). The standard CNN has 4 convolutional layers (16–32–64–64 channels) followed by adaptive average pooling and a Linear(1024, 128)–ReLU–Linear(128, 10) classifier.

F.4 Aggregation procedures

PCam. The Pareto analysis (Figure 2, Table 6) uses a single seed (42) to reduce computational cost across the full growth-rate sweep. The data efficiency comparison (1% vs. 10%) uses 3 seeds per configuration.

OrganMNIST3D. All results in Table 7 report mean $\pm$ std over 3 seeds (42, 123, 456). Rotation robustness (σ_{rot}) is the standard deviation of test accuracy across all 24 octahedral rotations of the test set. The data efficiency analysis (Table 5 in main text) uses 3 seeds per data fraction.

Spherical MNIST. All results in Table 8 report mean $\pm$ std over 3 seeds (42, 123, 456). Rotation robustness (σ_{rot}) is measured by applying 10 random $\mathrm{SO}(3)$ rotations to the entire test set and computing the standard deviation of accuracy, yielding 30 measurements per model (10 rotations $\times$ 3 seeds).

F.5 Dataset licenses

- **PCam**: CC0 1.0 Universal (public domain).
- **OrganMNIST3D**: CC BY 4.0.
- **Spherical MNIST**: derived from MNIST, which is publicly available for research use.

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